



Buffet Working Group Summary Presentation



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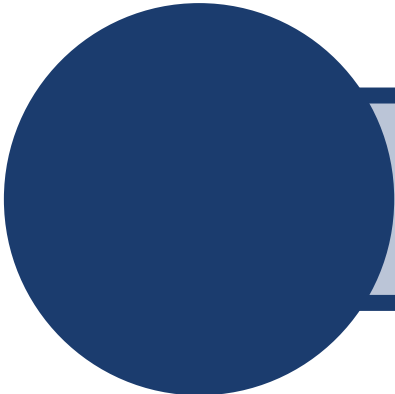
NASA Langley Research Center
Aeroelasticity Branch

Andrea Sansica

JAXA
Fundamental Technology Research Unit
Aviation Technology Directorate



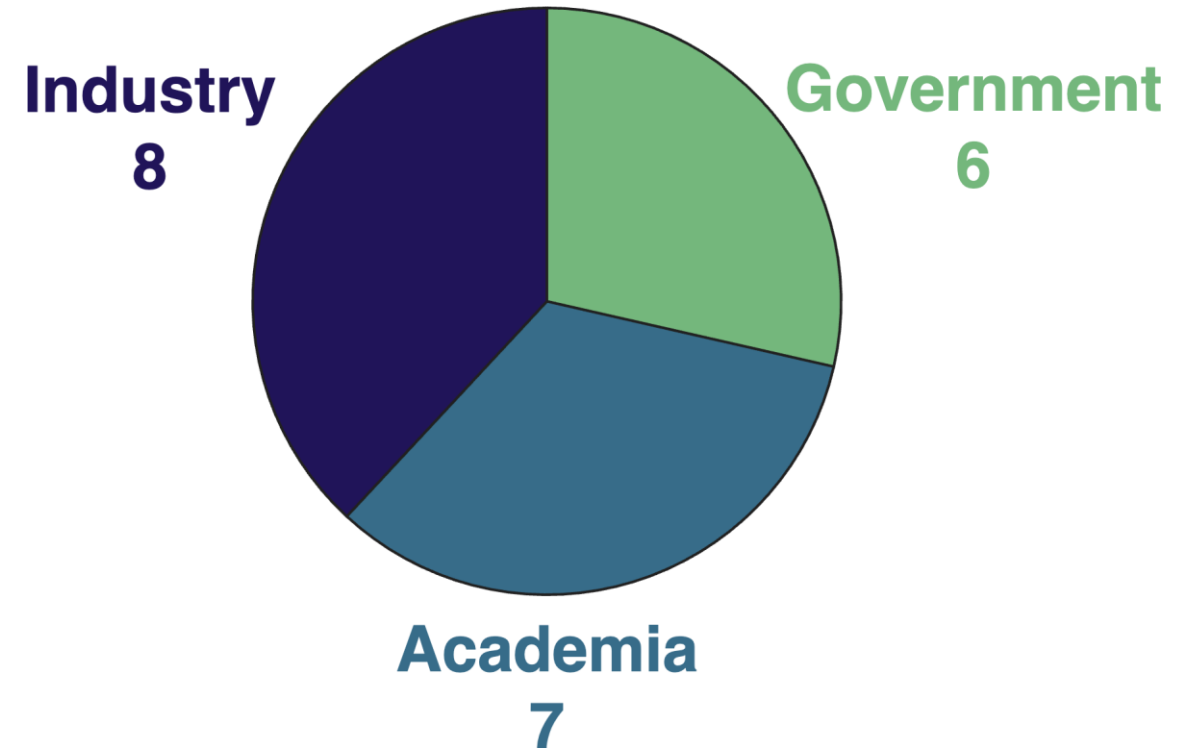
- **Background and Participants**
- **Test Case 1**
- **Test Case 2**
- **Test Case 3**
- **Desired Future Work**

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Background and Participants

- **Organizing Committee**
 - Hadar Ben-Gida, Technion-Israel Institute of Technology
 - Brent Pomeroy, NASA Langley Research Center (Emeritus)
 - Daniela Raveh, Technion-Israel Institute of Technology
 - Andrea Sansica, JAXA
 - Bret Stanford, NASA Langley Research Center
- **Test Case 3 was one of the motivations for joint workshop**
 - Test Cases 2 and 3 matured quickly
 - Goals largely crafted before DPW-8/AePW-4 launch (SciTech 2024)

- **21 teams from varied agency types**
 - Industry: 8
 - Government: 6
 - Academia: 7
- **Large group**
 - 104 on email distribution list
 - 15-30 participants per call
- **Monthly tagups**
- **Strong presence at mini workshops**



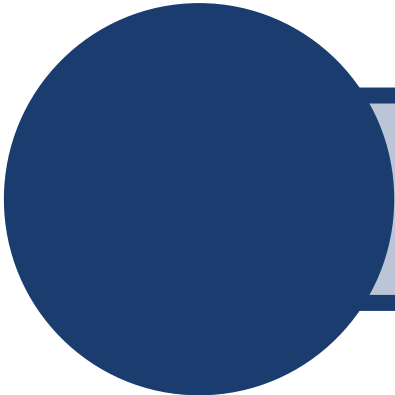
- **A retrospective of DPW learnings is appropriate**
- **The following summary indicates general trends and opinions for applied transonic wing/airfoil aerodynamics**
 - These are general, debatable, statements from DPW learnings
 - The chart is not intended to be comprehensive
- **Corrections and tuning can be applied to each of these turbulence models**
- **Characteristics are subject to spirited discussion**

DPW Turbulence Model Learnings: An Attempt

Model	Type	Characteristics
SA		
k- ω		
k- ϵ		
SST		
RSM		

DPW Turbulence Model Learnings: An Attempt

Model	Type	Characteristics
SA	• Solves for modified kinematic eddy viscosity	• Established dependability with well-behaved flows including adverse pressure gradients and mild/moderate boundary layer separation • Separation may be delayed
k- ω	• Solves for kinetic energy and specific dissipation rate, indirectly capturing eddy viscosity	• Sensitive to freestream conditions and outer flow contamination
k- ϵ	• Solves for kinetic energy and dissipation rate, indirectly capturing eddy viscosity	• Can delay or miss separation, especially due to strong adverse pressure gradients
SST	• Includes considerations of k-ω and k-ϵ • Modifies eddy viscosity to transport turbulent shear stress	• Established dependability with well-behaved flows including adverse pressure gradients and mild/moderate boundary layer separation • May yield early separation
RSM	• Solves 7 equations, including full turbulent stress tensor • Accounts for anisotropic turbulence	• High-curvature flowfields can be captured well • Slower convergence and greater numerical stiffness



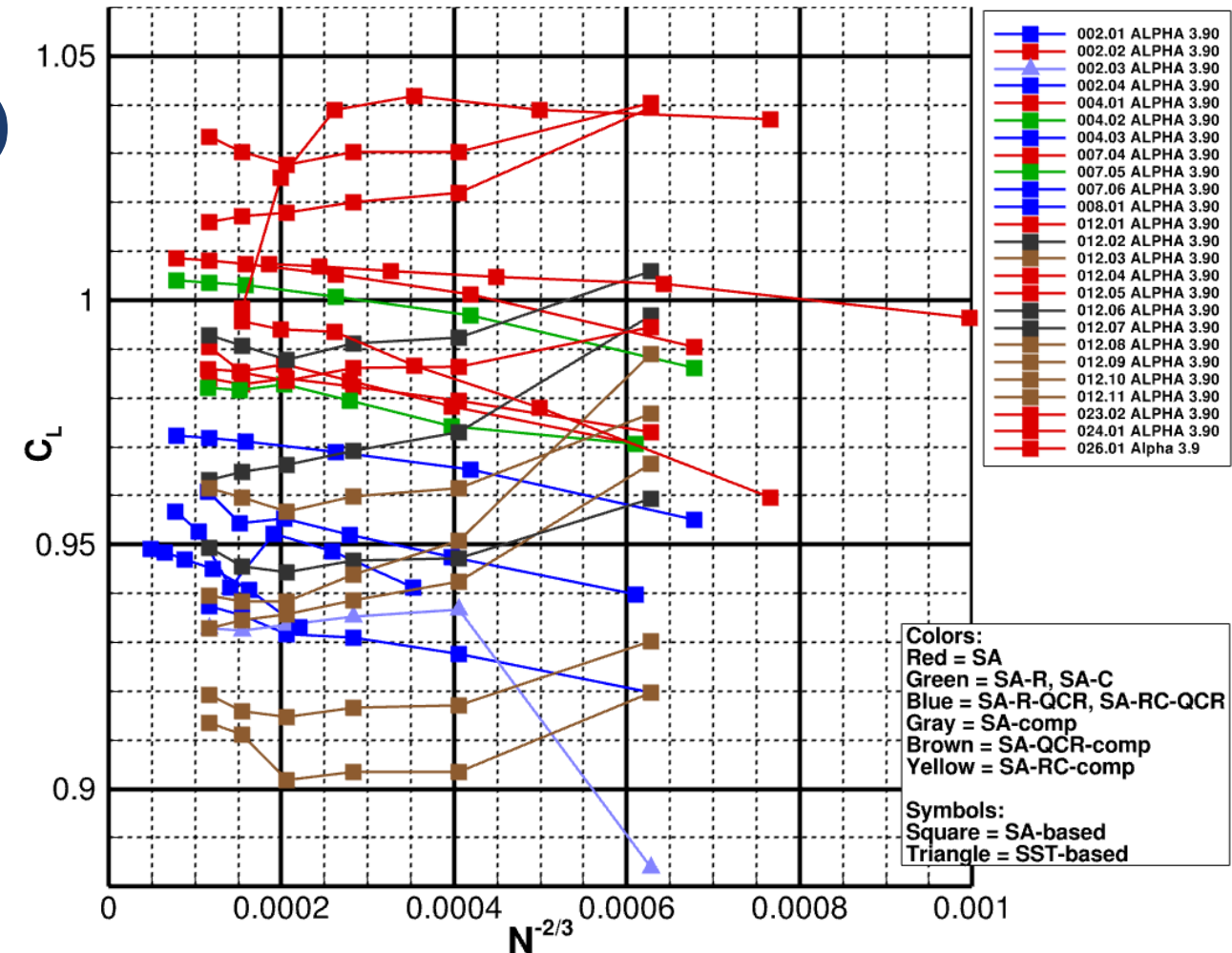
Test Case 1

- **Many custom grids were generated**
 - Few passed export control and thus grid comparisons were not possible
 - Wide variety of gridding practices and approaches
- **Significant student participation**
- **Experimental reference:**

Jaquin, et al. "Experimental Study of Shock Oscillation over a Transonic Supercritical Profiles." AIAA Journal, Vol. 47, No. 9, 2009. Pages 1985-1994.

Test Case 1a: Grid Level Effect at 3.90 deg

- Results shown at 3.9 deg, but representative of high-alpha cases
- SA, SA-R, SA-C (“SA,” red and green)
 - Elevated C_L
 - Elevated C_D
 - Decreased C_M
- SA-QCR, SA-comp, SA-QCR-comp, SA-RC-comp (“QCR/comp,” blue, black, brown)
 - Decreased C_L
 - Decreased C_D
 - Elevated C_M
- Few contributions with SST; no RSM (used in Test Case 2)

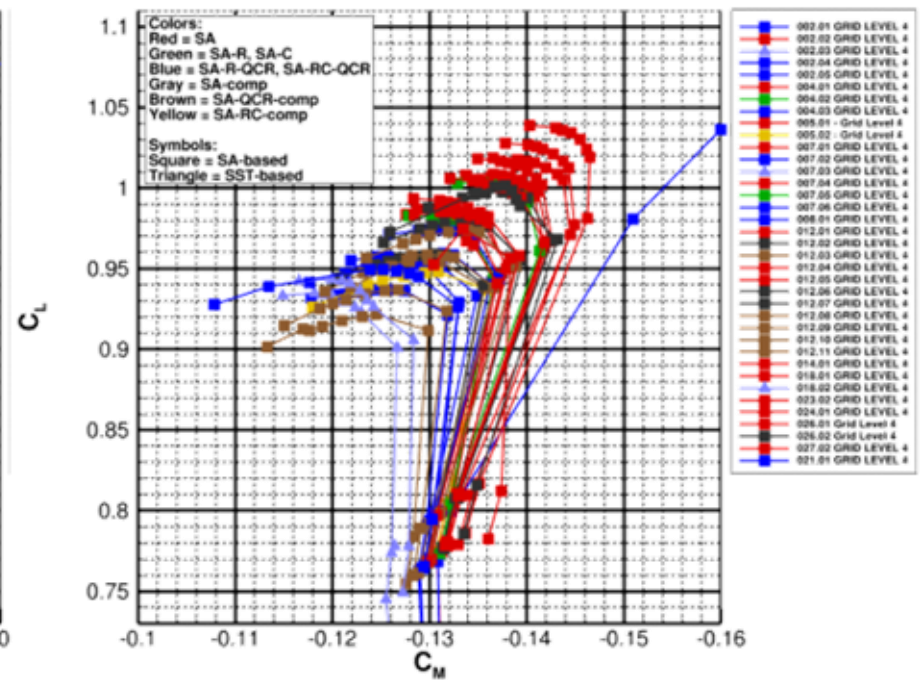
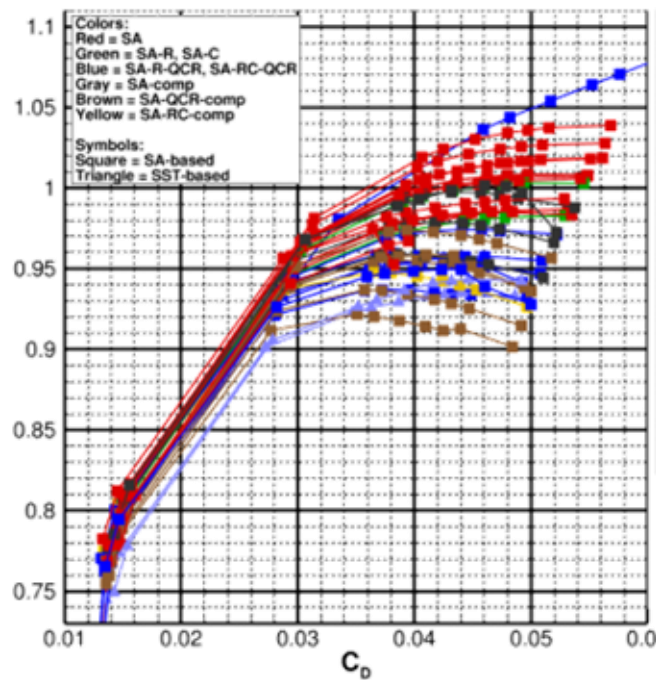
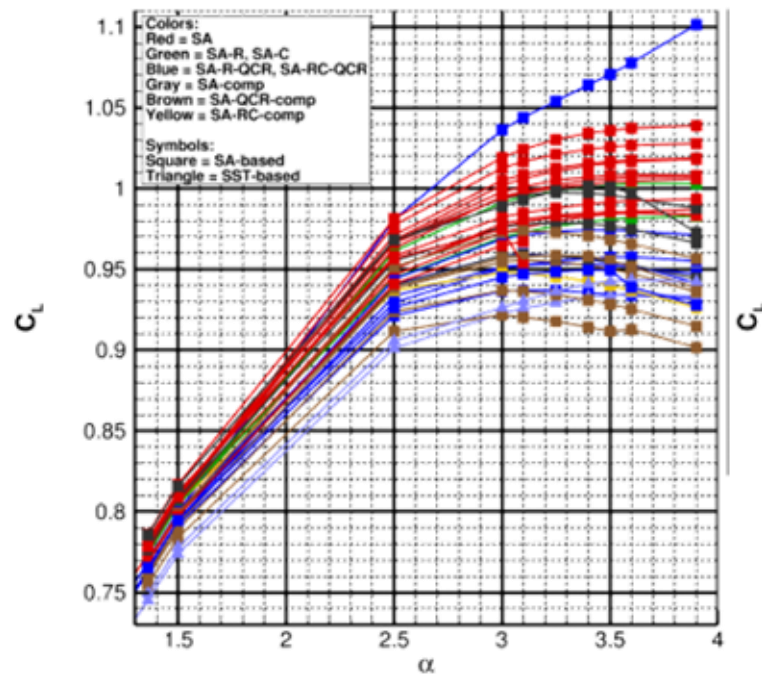


Test Case 1a: Integrated F&M

- Increased scatter relative to DPW-VII
- SA group: red and green
 - Elevated C_L , elevated C_D , decreased C_M
- QCR-comp group: blue, gray, and brown
 - Decreased C_L , decreased C_D , increased C_M

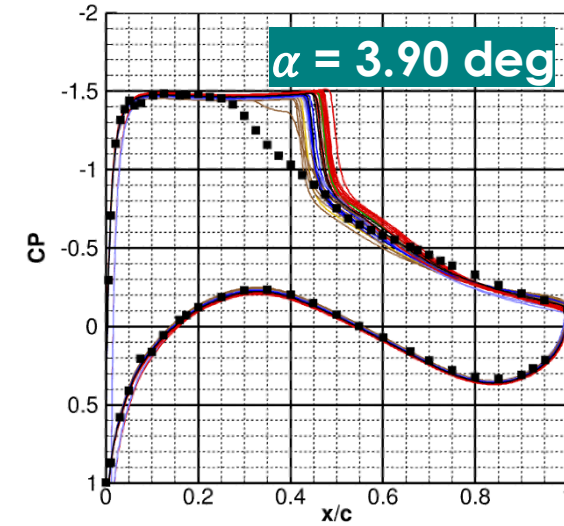
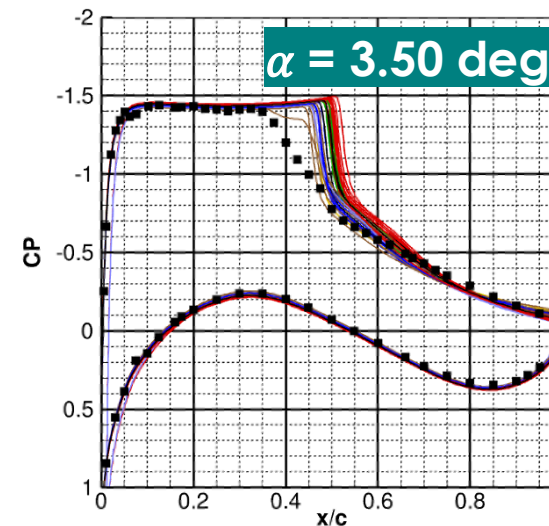
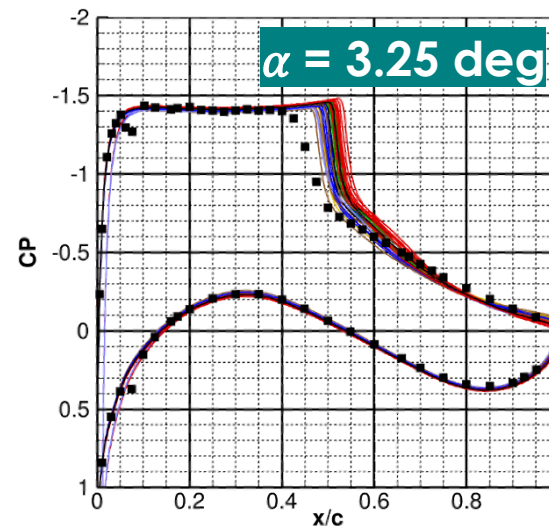
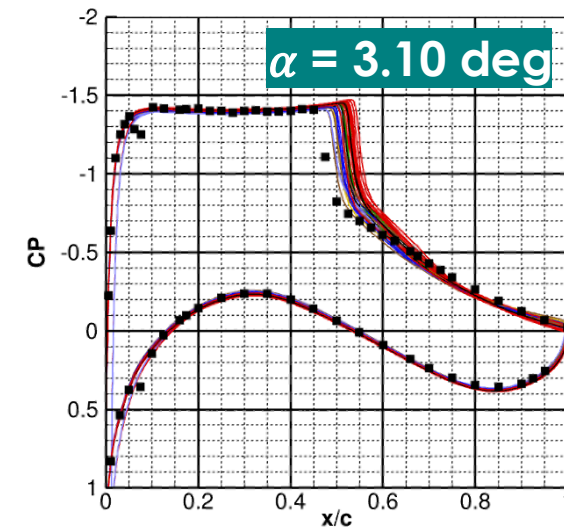
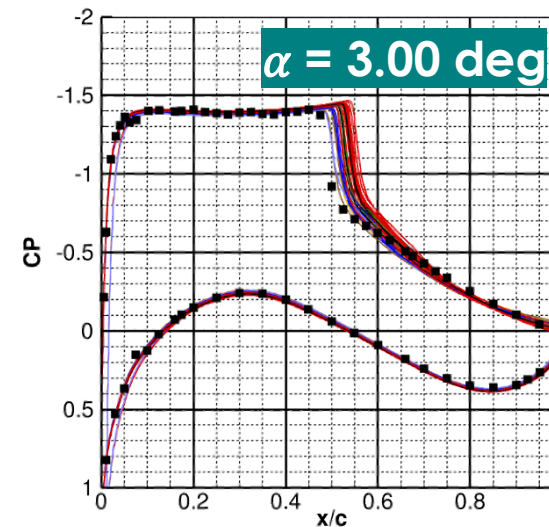
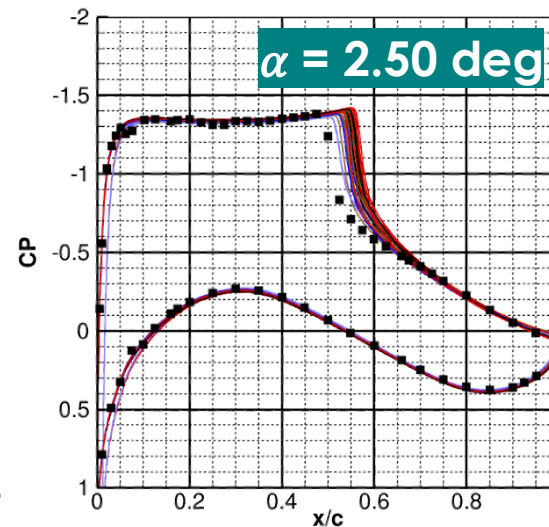
SA
 SA-R, SA-C
 SA-R-QCR, SA-RC-QCR
 SA-comp
 SA-QCR-comp
 SA-RC-comp

Squares: SA
 Triangles: SST



Test Case 1a: Sectional Cuts

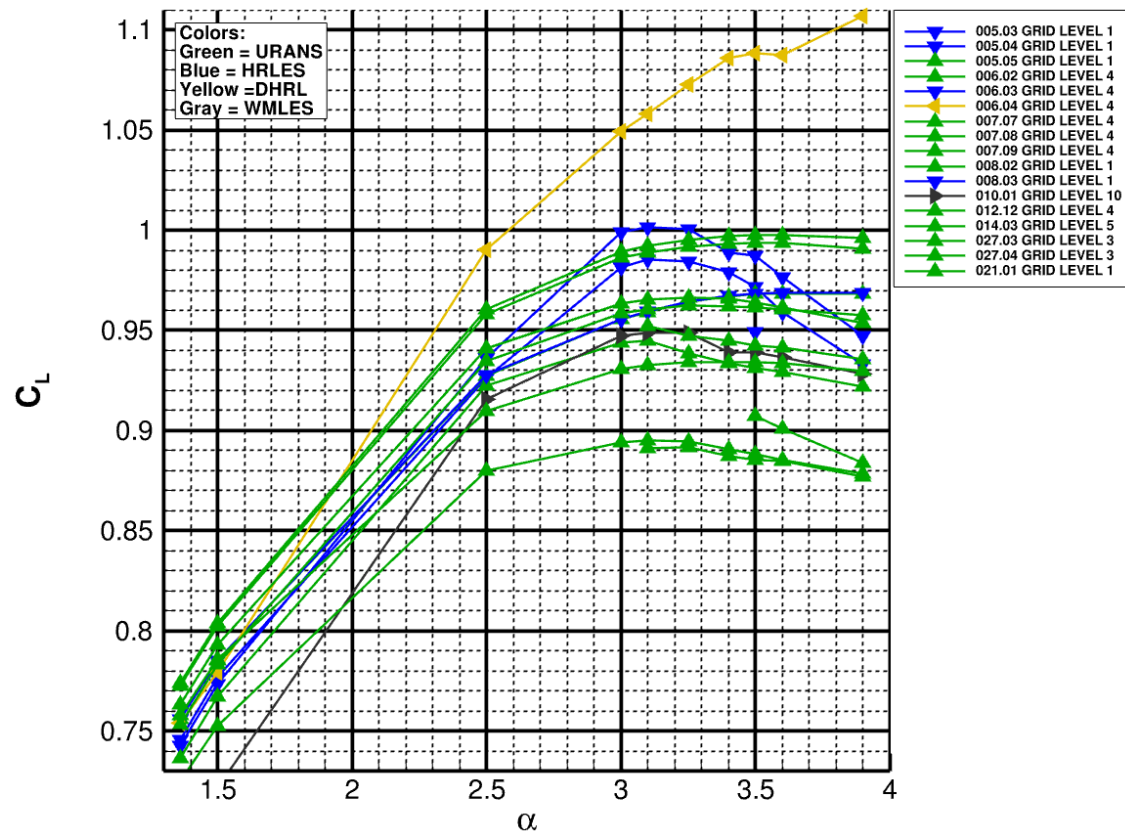
- Most solutions predict the shock too far downstream
 - “-comp” correction gives improvements
- **Steady calculations**
 - Shock does not move
 - Simulations do not capture the smooth (time-averaged) gradient across the shock for post-onset cases



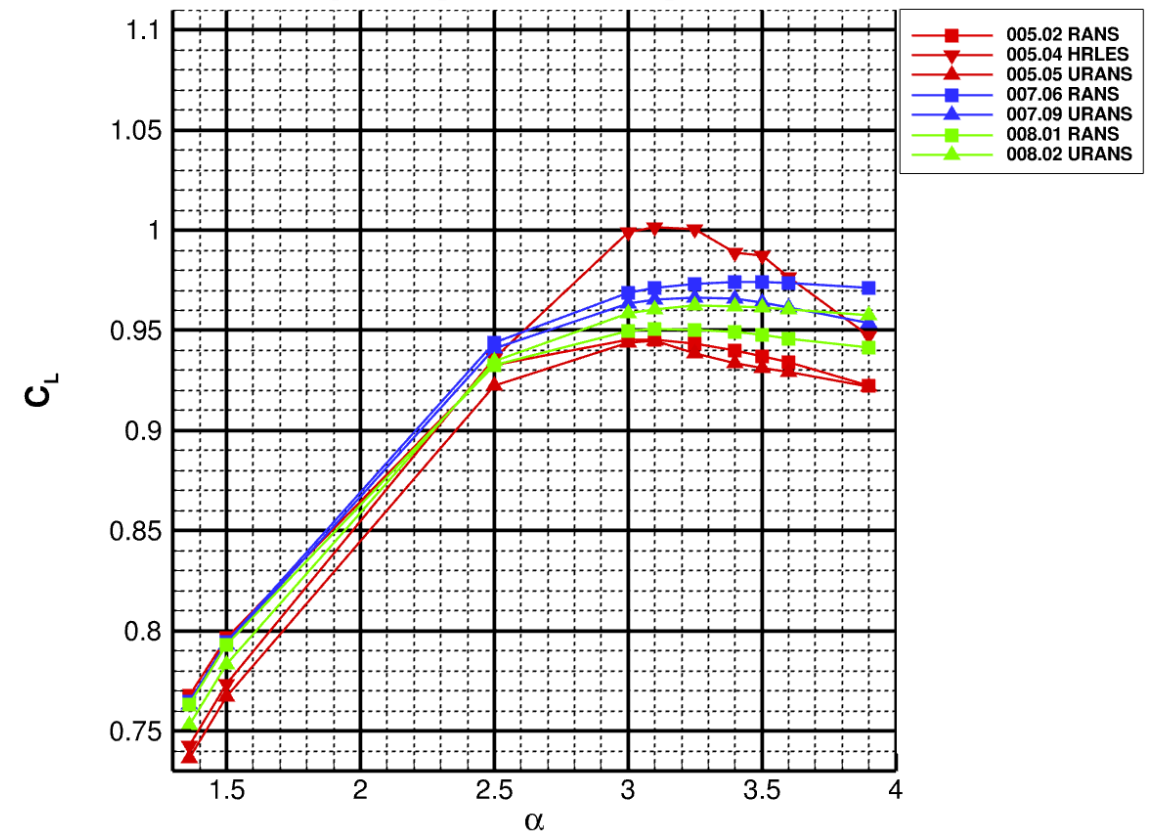
Test Case 1b: Integrated F&M

- For the same setup, steady & unsteady values do not match pre-buffet
- Early RANS separation anticipated but not necessarily observed

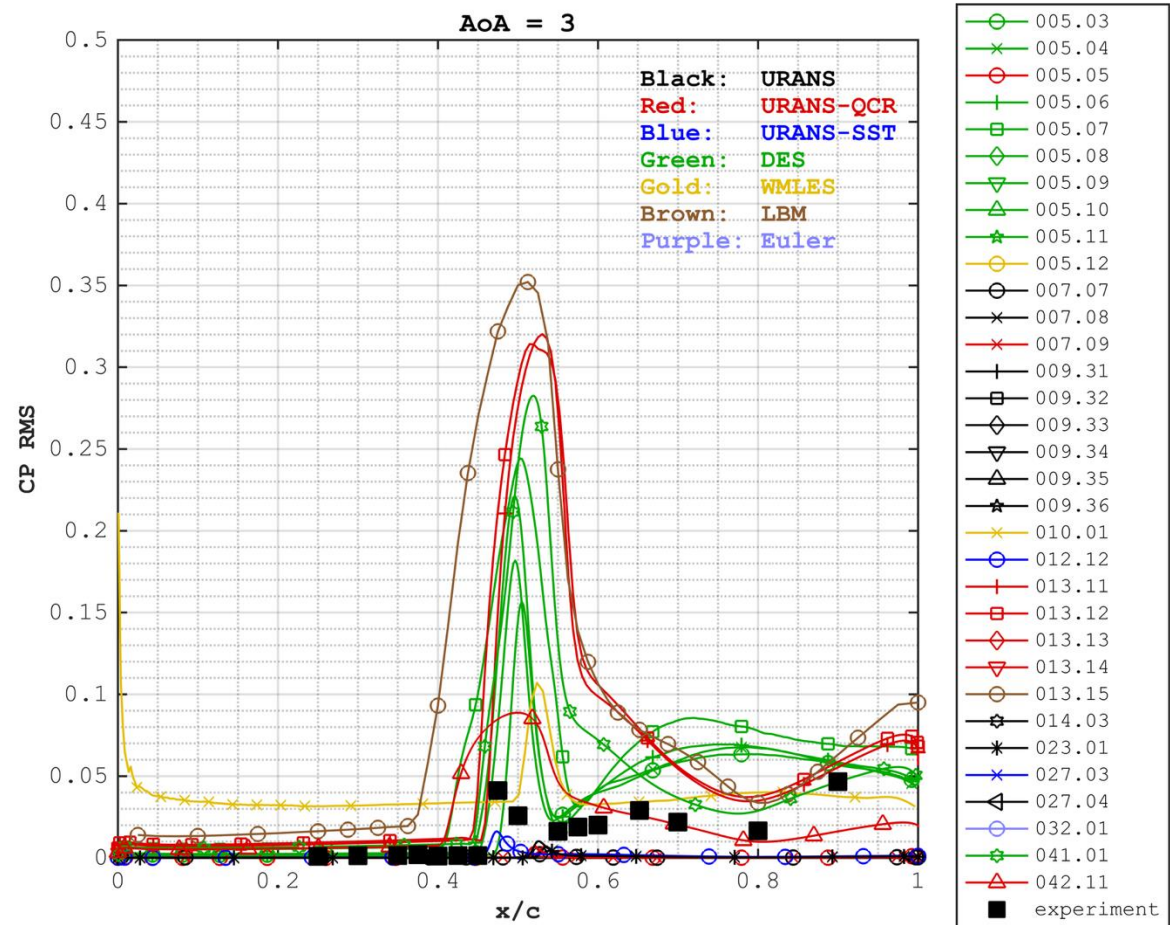
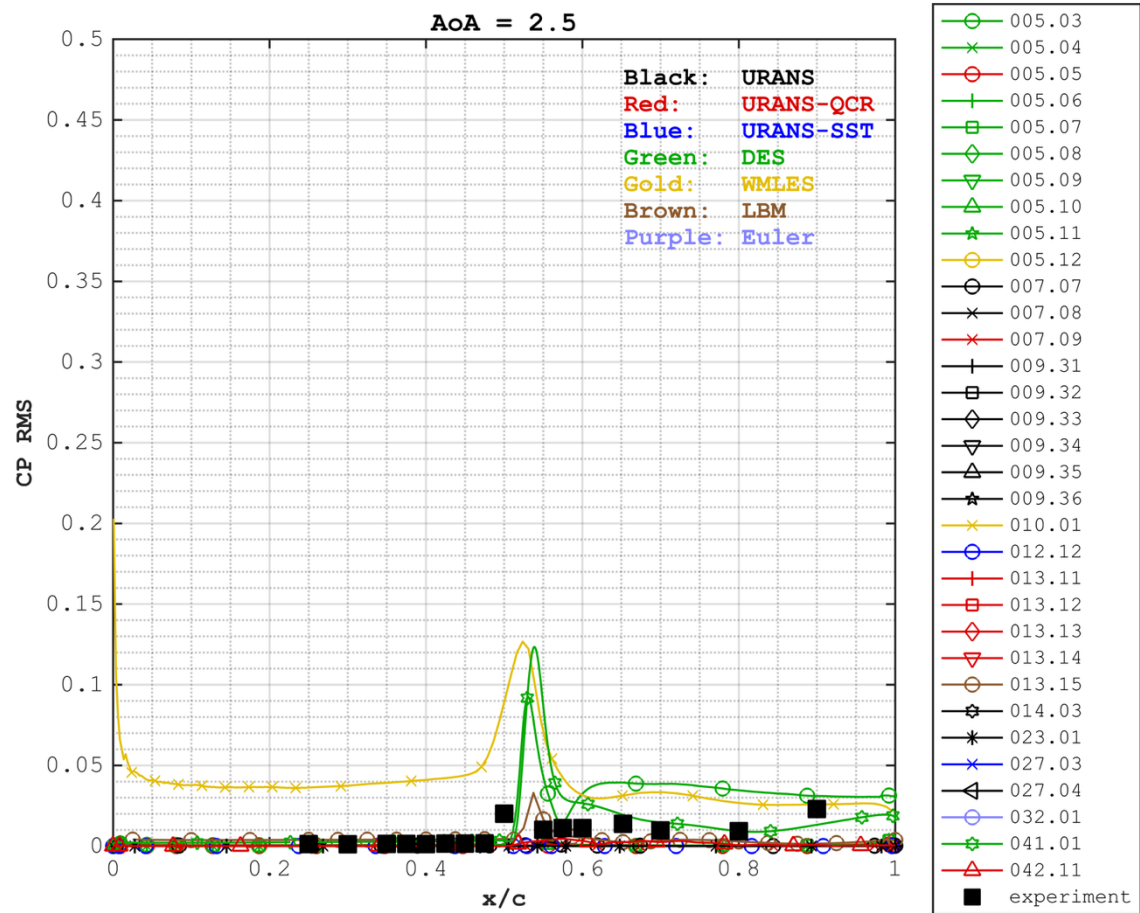
Colored by method



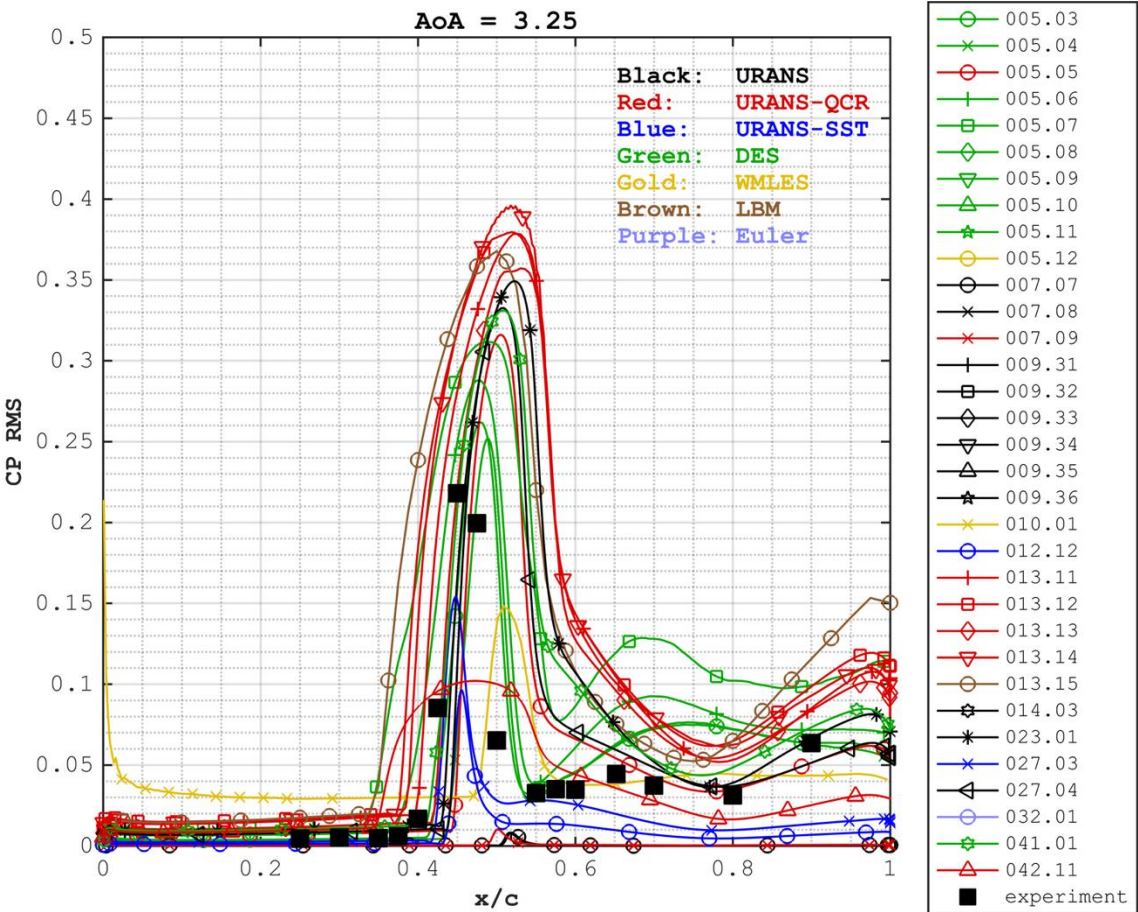
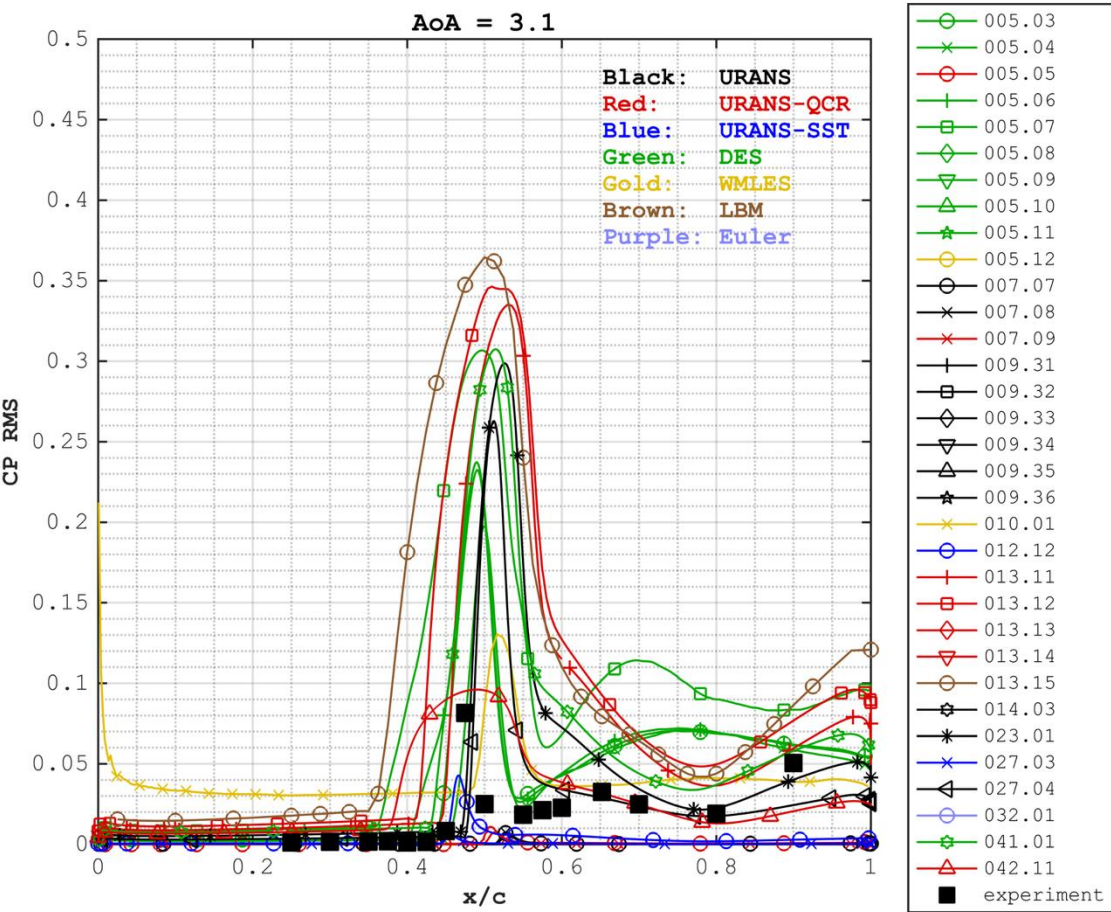
Colored by ID (selected) contributions



Unsteady Sectional Cuts: Pre-Bufferet Onset

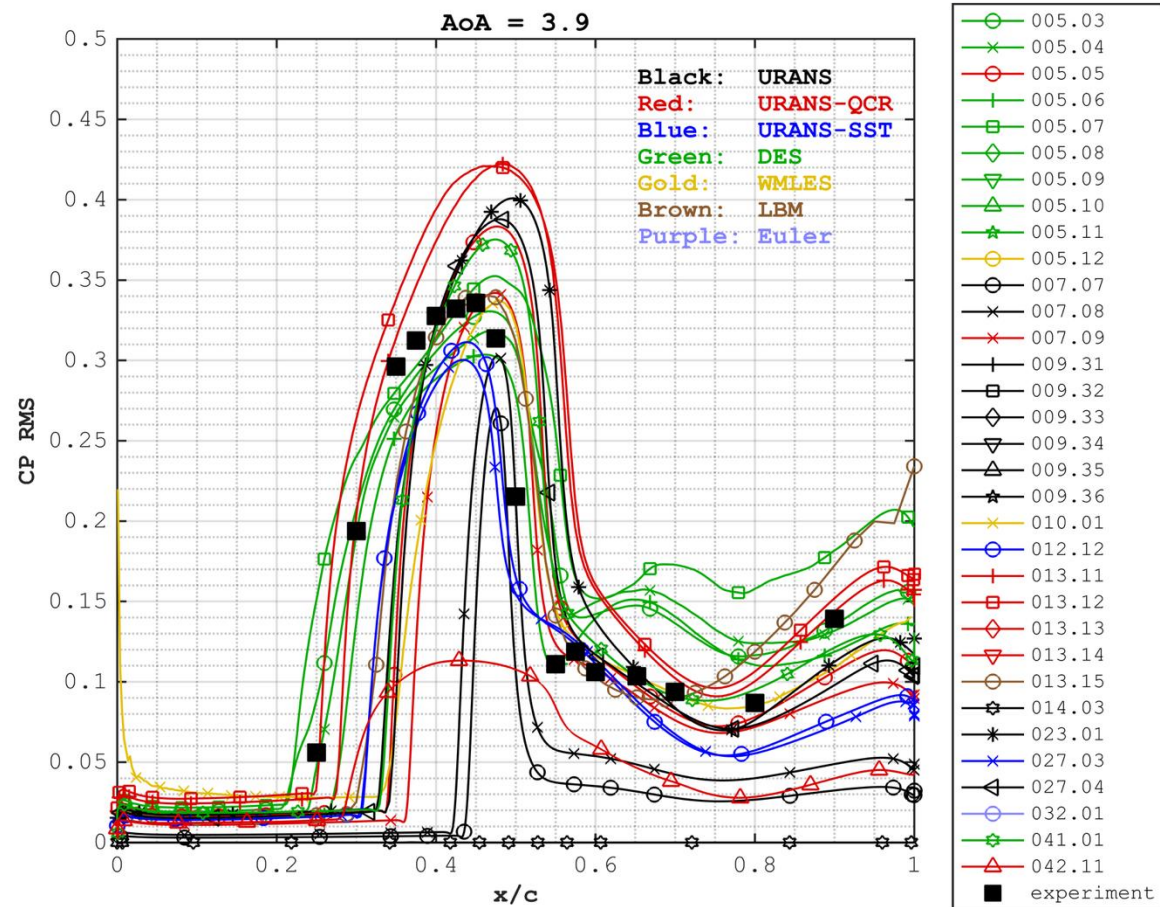
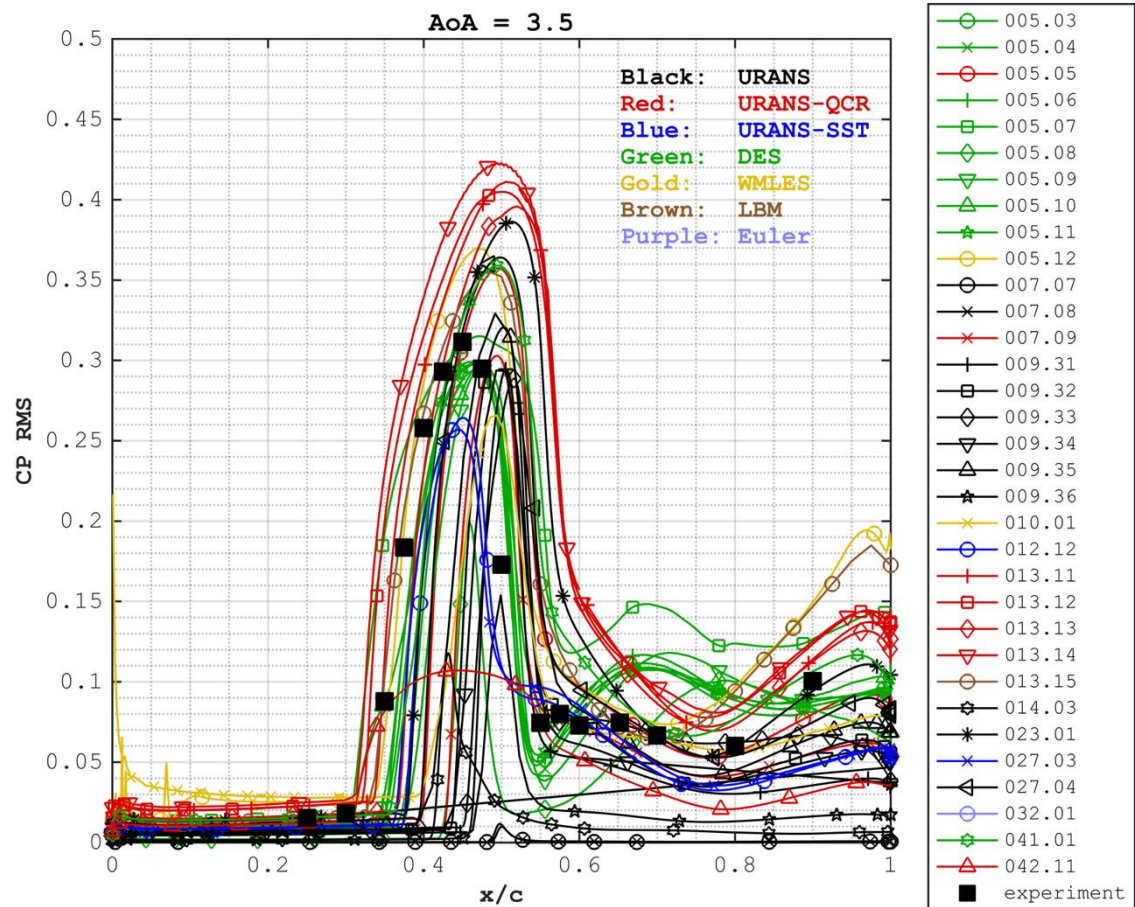


Unsteady Sectional Cuts: Post-Buffer Onset

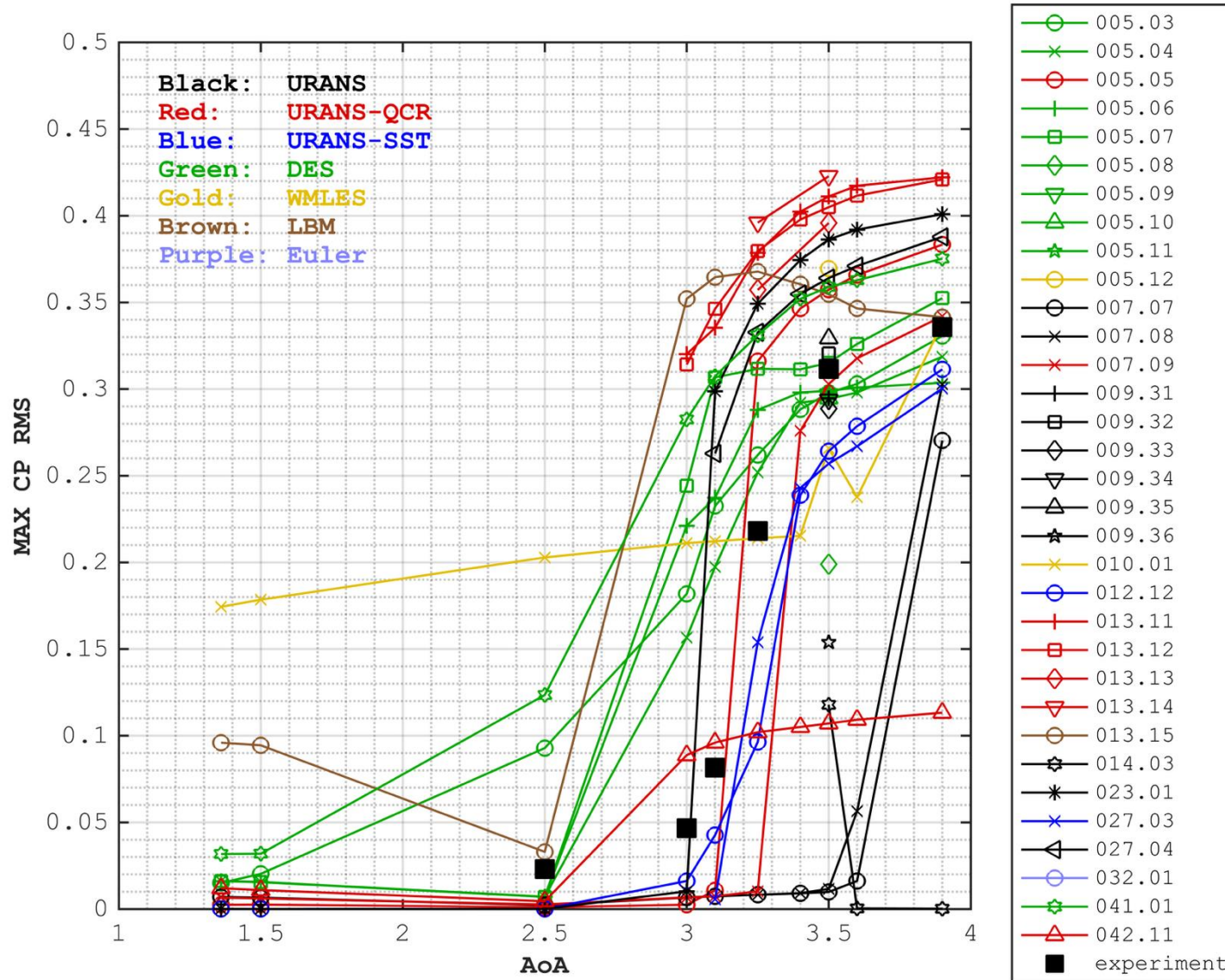


Note: Buffet onset measured at 3.1 deg in Jaquin, et al.

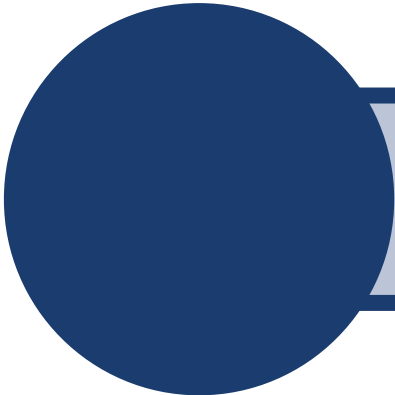
Unsteady Sectional Cuts: Post-Buffer Onset



Unsteady Sectional Cuts: AoA Sweep



- The SST-based solvers are closest to the experimental data, in terms of both buffet onset and post-buffet RMS levels
 - Only two SST submissions, however
- Most other results (QCR-based, DES) predict early buffet onset by 0.5-1 deg and overpredict RMS levels
- No RSM

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Test Case 2

- **Parametric sweeps presented showing F&M and sectional cuts**
 - Temporal scheme
 - Turbulence model
 - Grid type
 - Grid level
- **Surface contour plots are shown at the end of the section**
- **No zeroing of data to a consistent point (e.g., 2.29 deg)**
 - Need universal simulations at 2.29 deg
 - Sometimes called the Tinoco correction

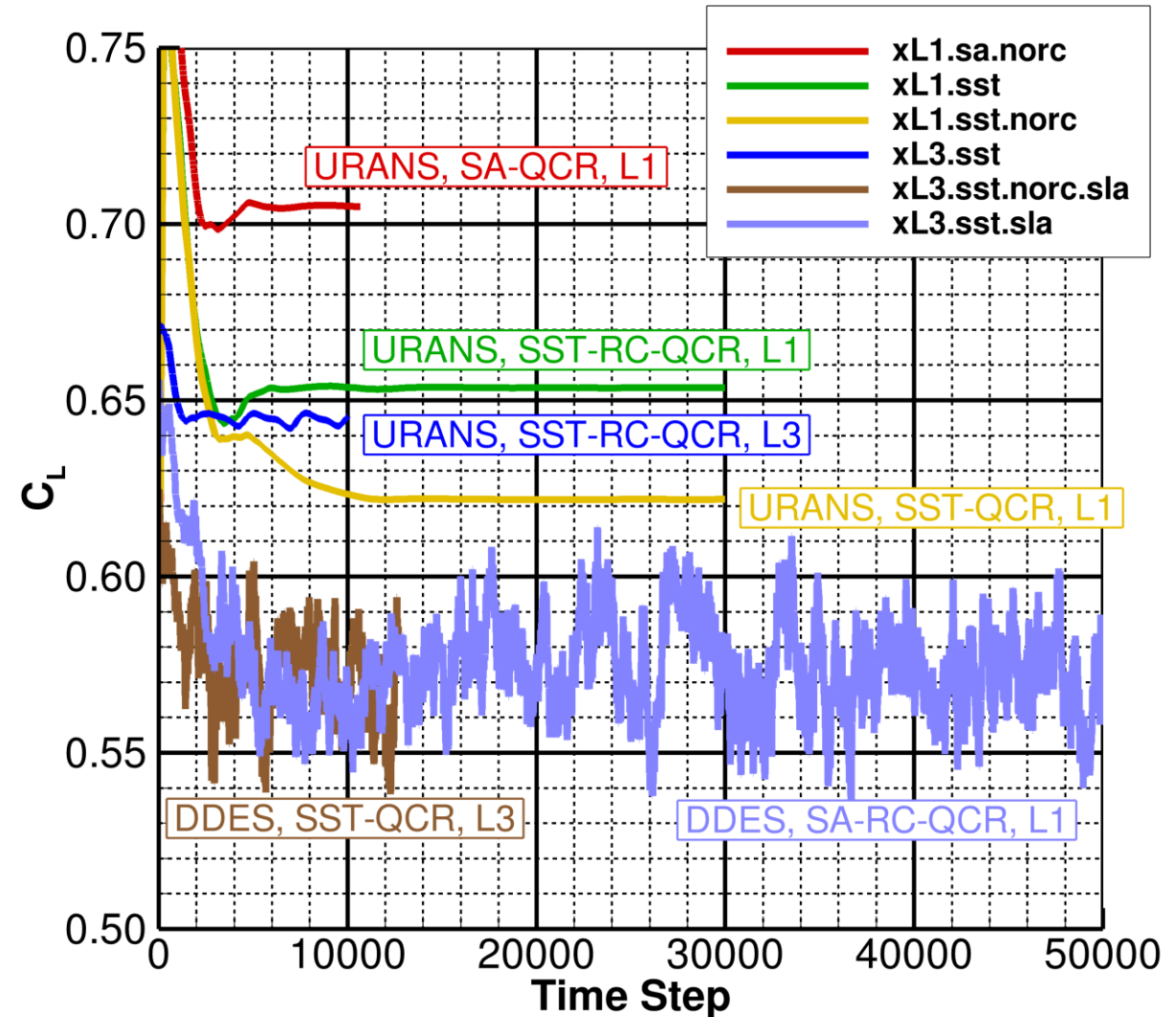
- **Experimental reference:**

Shunsuke Koike, Makoto Ueno, Kazuyuki Nakakita and Atsushi Hashimoto, "Unsteady Pressure Measurement of Transonic Buffet on NASA Common Research Model", AIAA 2016-4044, June 2016.

Temporal Formulation	Turbulence Model	Grid	Submission ID
URANS	SA-RC-QCR	Helden	007.10
	SA-RC-QCR	Helden	007.11
	SA-RC-QCR	Cadence	007.12
	SA-RC-QCR	Cadence	007.13
	SST	Cadence	021.02
	RSM	Cadence	027.05
HRLES	SA-neg	Ames Multiblock	005.01
	SST	Cadence	021.03
	SA-neg	Cadence	041.02
Lattice Boltzman	k- ϵ	Custom Cartesian	013.01
	k- ϵ	Custom Cartesian	013.02
WMLES	Dynamic Smagorinsky	Custom Hex	002.01
	Dynamic Smagorinsky	Custom Vorni	010.02
Adaptive Euler	N/A	Custom Mixed Element	032.01

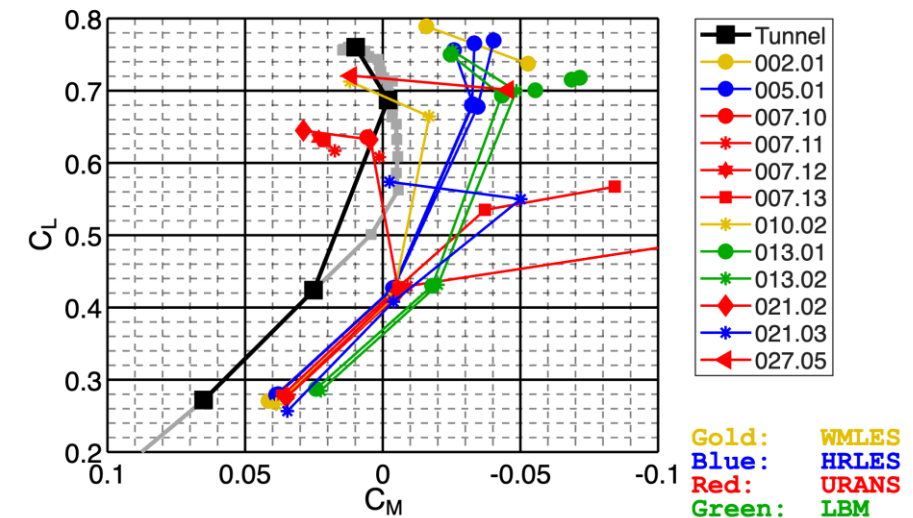
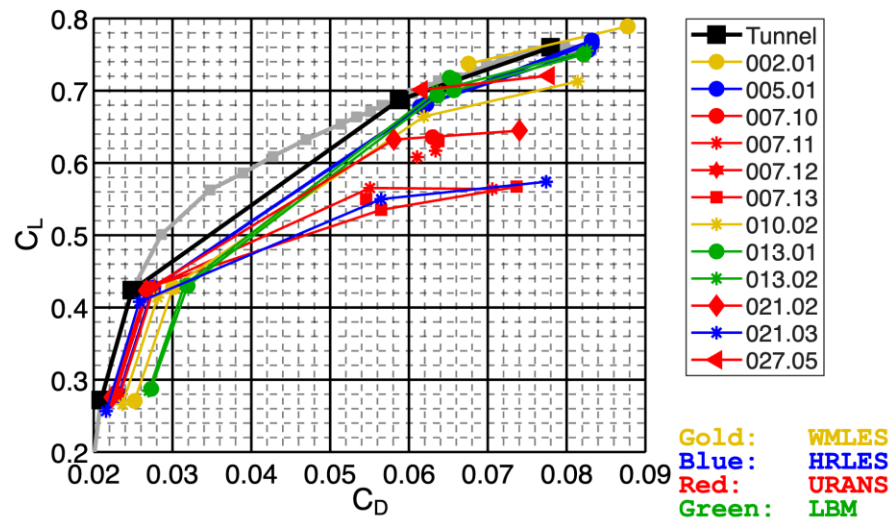
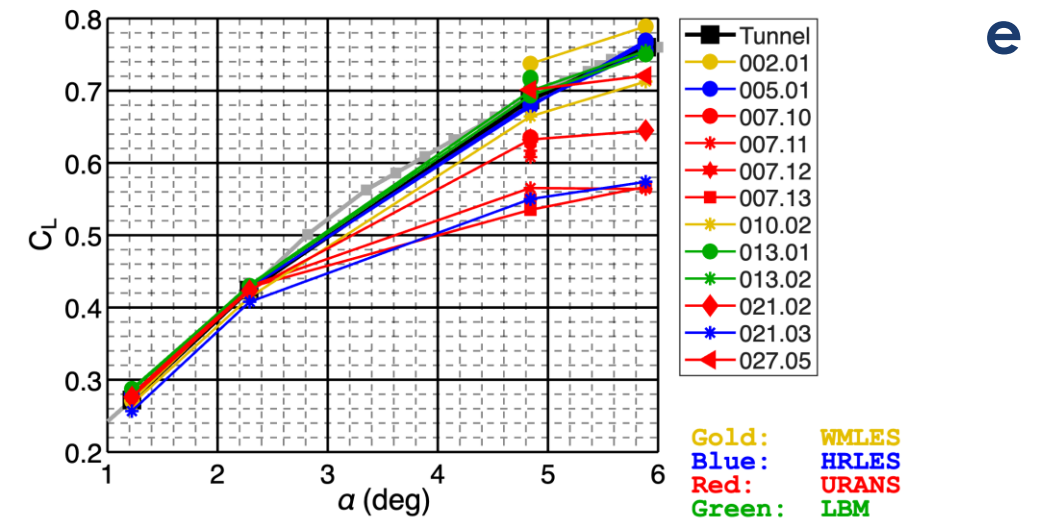
Buffet: A Difficult Aerodynamic Problem

- Time histories shown from one participant's initial investigation at 5.89 deg
- URANS may not yield unsteady flow for L1-L3
- Some participants had isolated success at L4+
- Determining truth can be difficult
- Additional work needed to understand time integration parameters



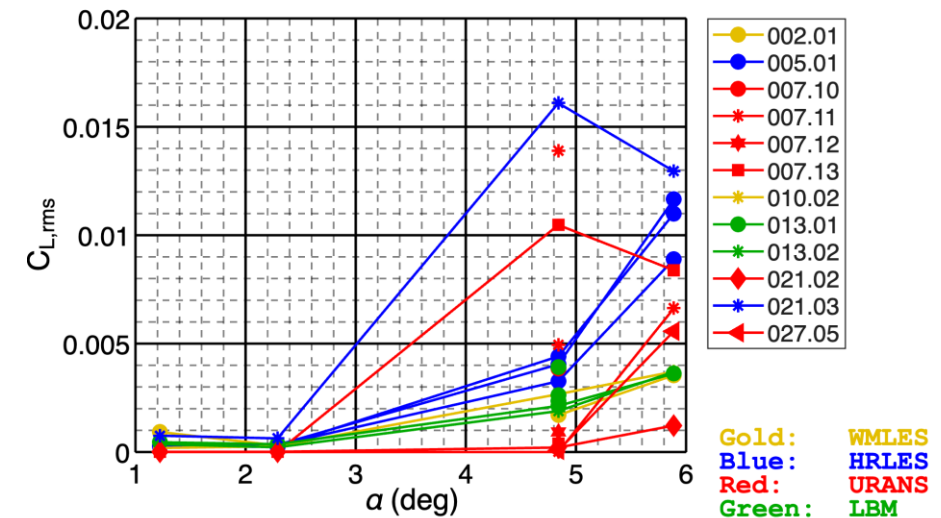
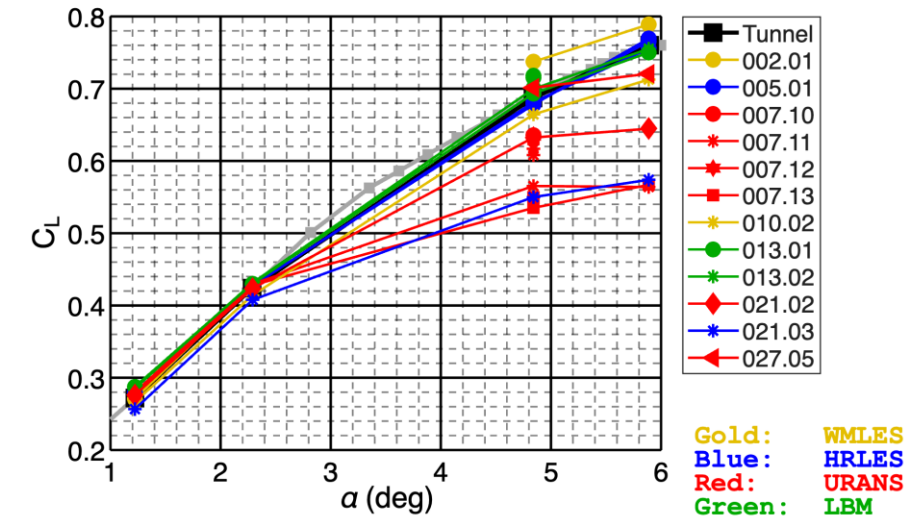
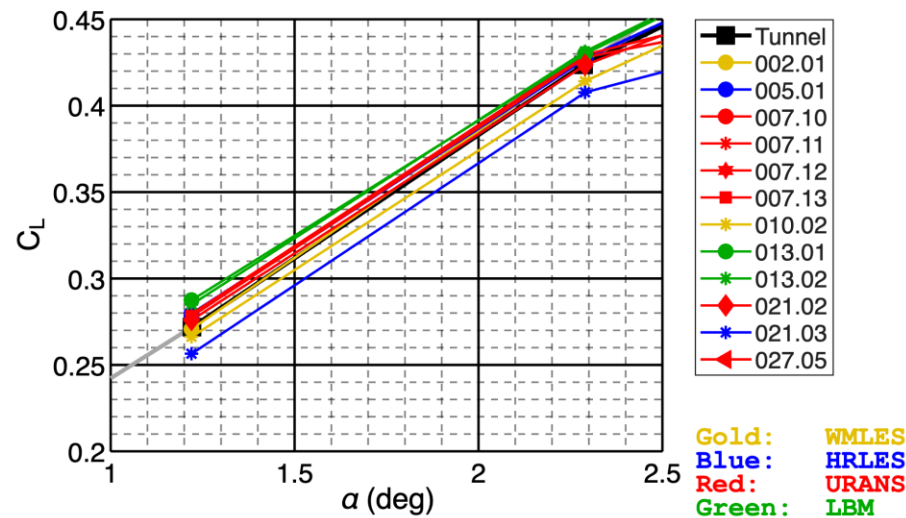
Temporal Formulation Effect – All F&M

- Encouraging clustering in lift and drag dependence model
- Elevated pitching moment scatter
- Multiple lines of same color/symbol are sub



Temporal Formulation Effect – Lift

- **Low-alpha**
 - C_L spread ± 0.02
 - Minimal $C_{L,rms}$ (no spurious unsteadiness), but some spread observed; URANS shows least
- **High-alpha**
 - URANS gives rapid drop in lift
 - Other schemes have some success
 - HRLES may yield fastest increase in $C_{L,rms}$



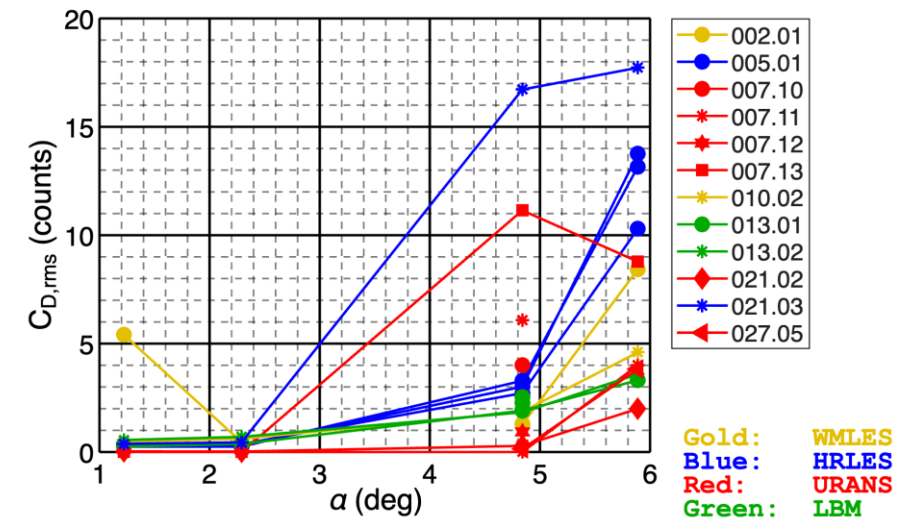
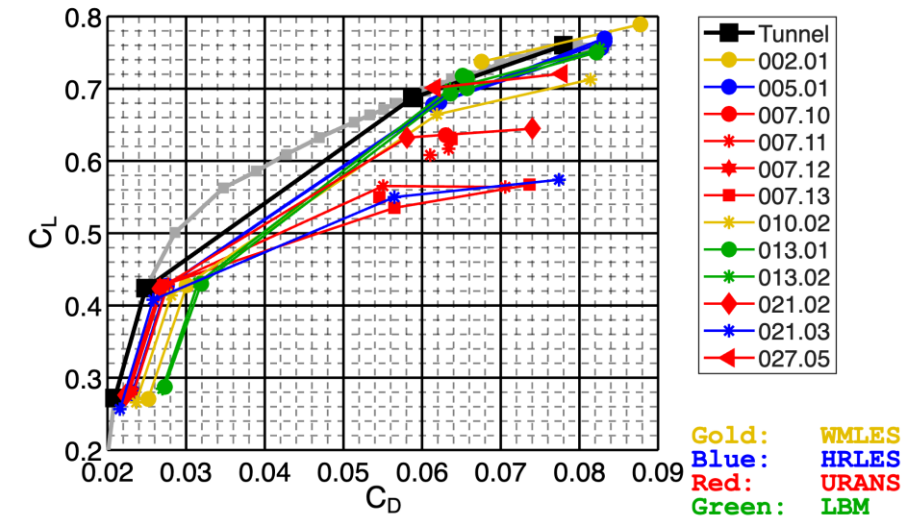
Temporal Formulation Effect – Drag

- **Low-alpha**

- Surprising scatter across all schemes (56.9 counts or 27.2%) at 2.29 deg
- Similar scatter for 1.24 and 2.29 deg
- WMLES/SD and LBM/k- ϵ yield highest drag
- Opportunity for an Unsteady Source of Scatter Working Group?
- WMLES may give spurious unsteadiness

- **High-alpha**

- URANS drop in C_L does not necessarily correspond to elevated C_D
- LBM yields least unsteadiness with HRLES indicating the most

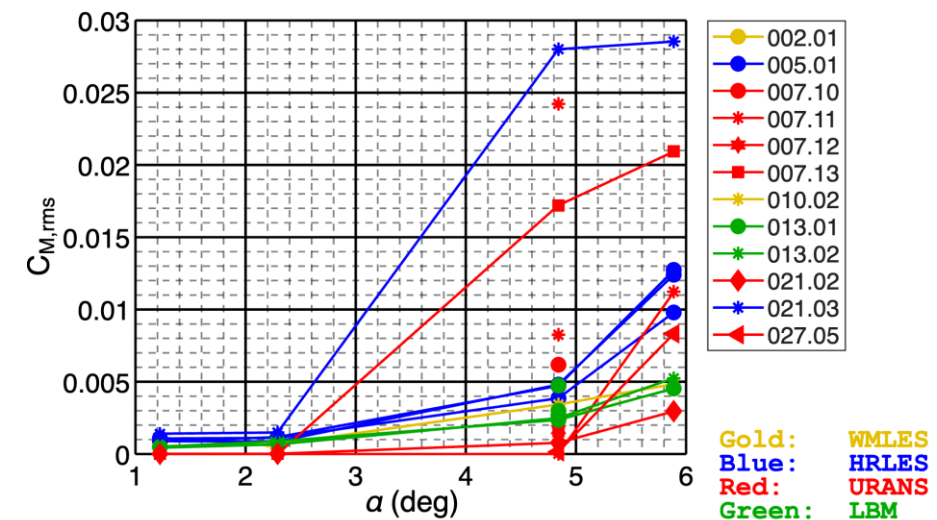
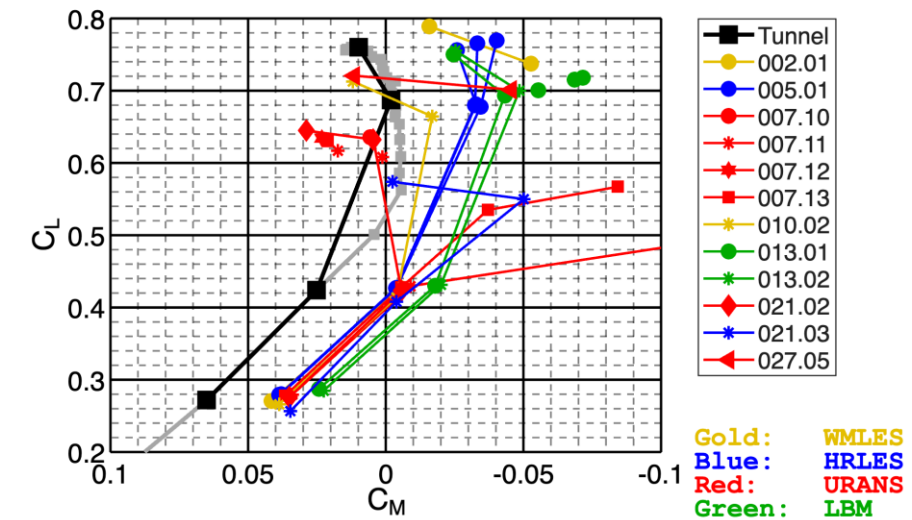


- **Low-alpha**

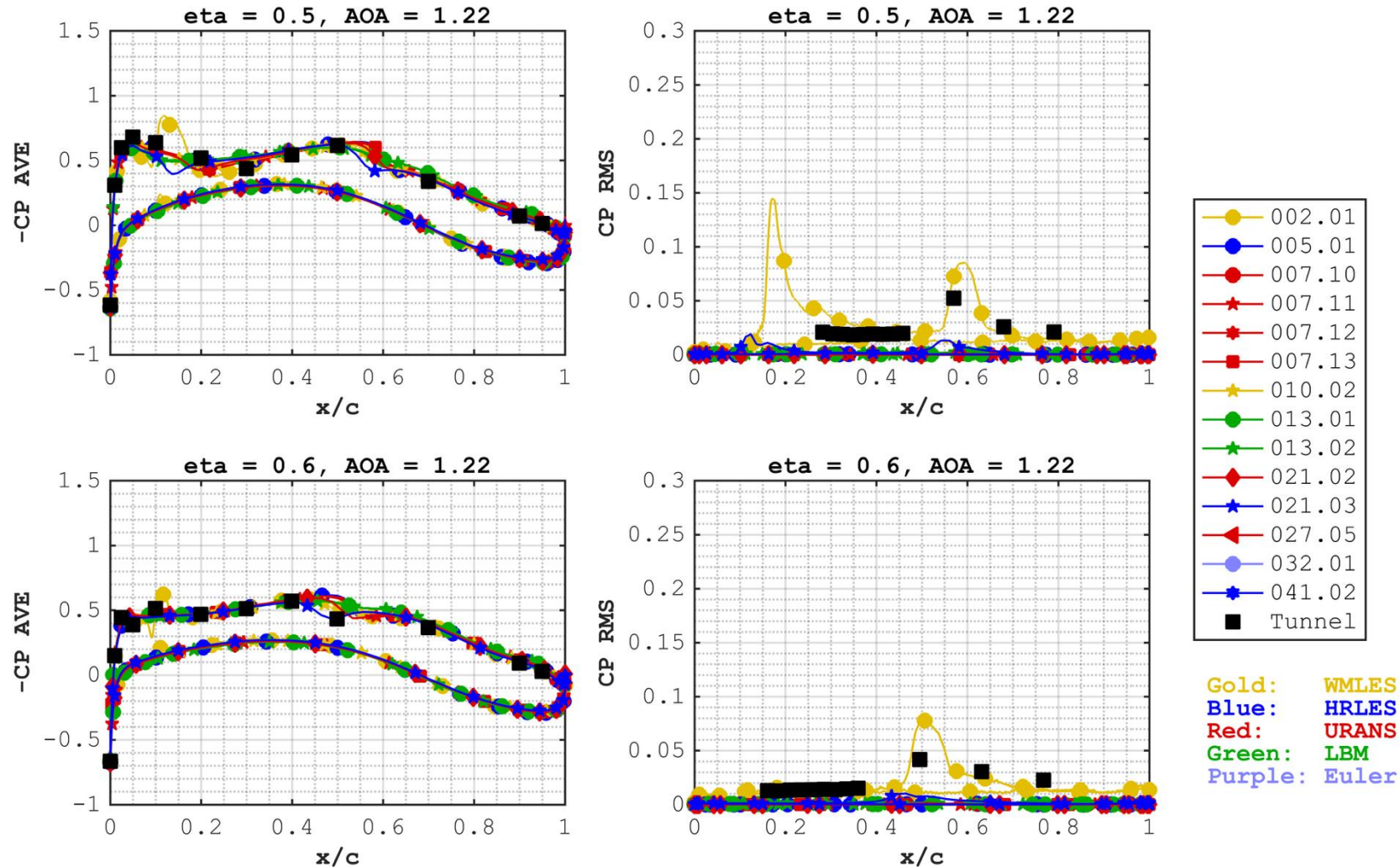
- Minimal $C_{M,rms}$ (consistent with $C_{L,rms}$)
- Consistent “rightward” shift of data relative to experimental data (not necessarily incorrect)

- **High-alpha**

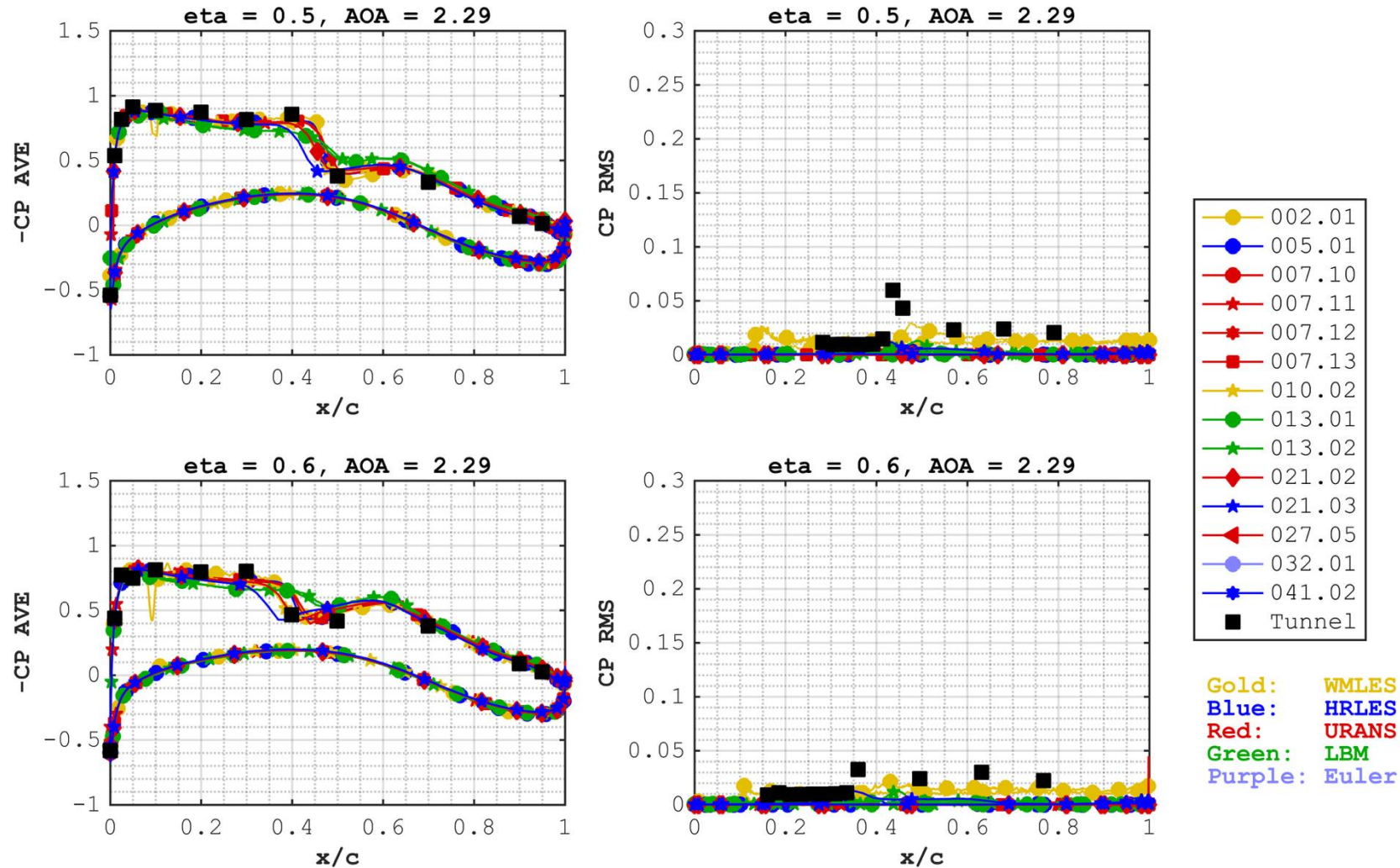
- URANS can predict large-magnitude nose-up pitching moment
- C_L - C_M shows similar scatter as DPW-VII, which was RANS-heavy (surprising)
- $C_{M,rms}$ trends mimic trends in both $C_{L,rms}$ and $C_{D,rms}$



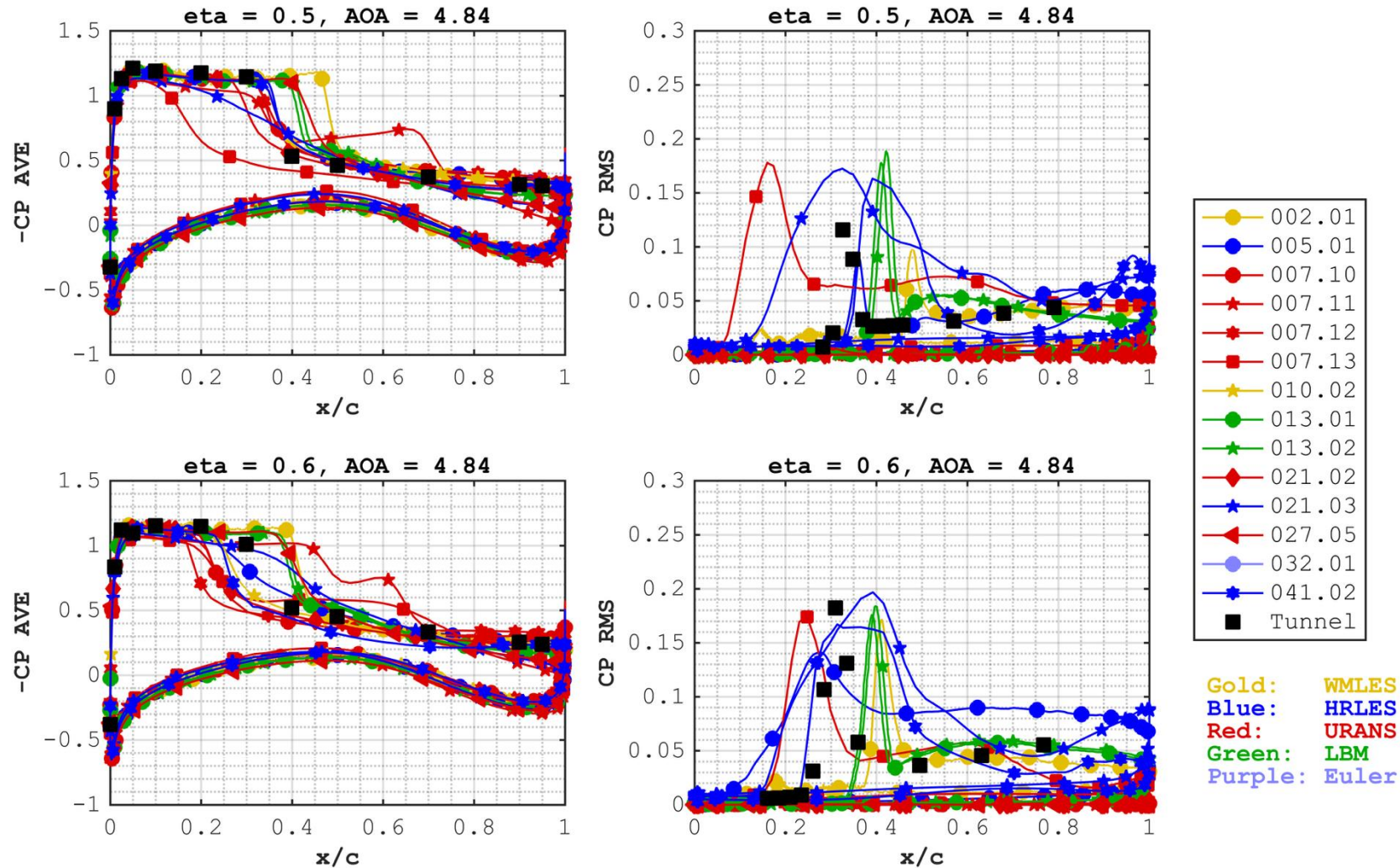
Temporal Formulation Effect – 1.22 deg



Temporal Formulation Effect – 2.29 deg

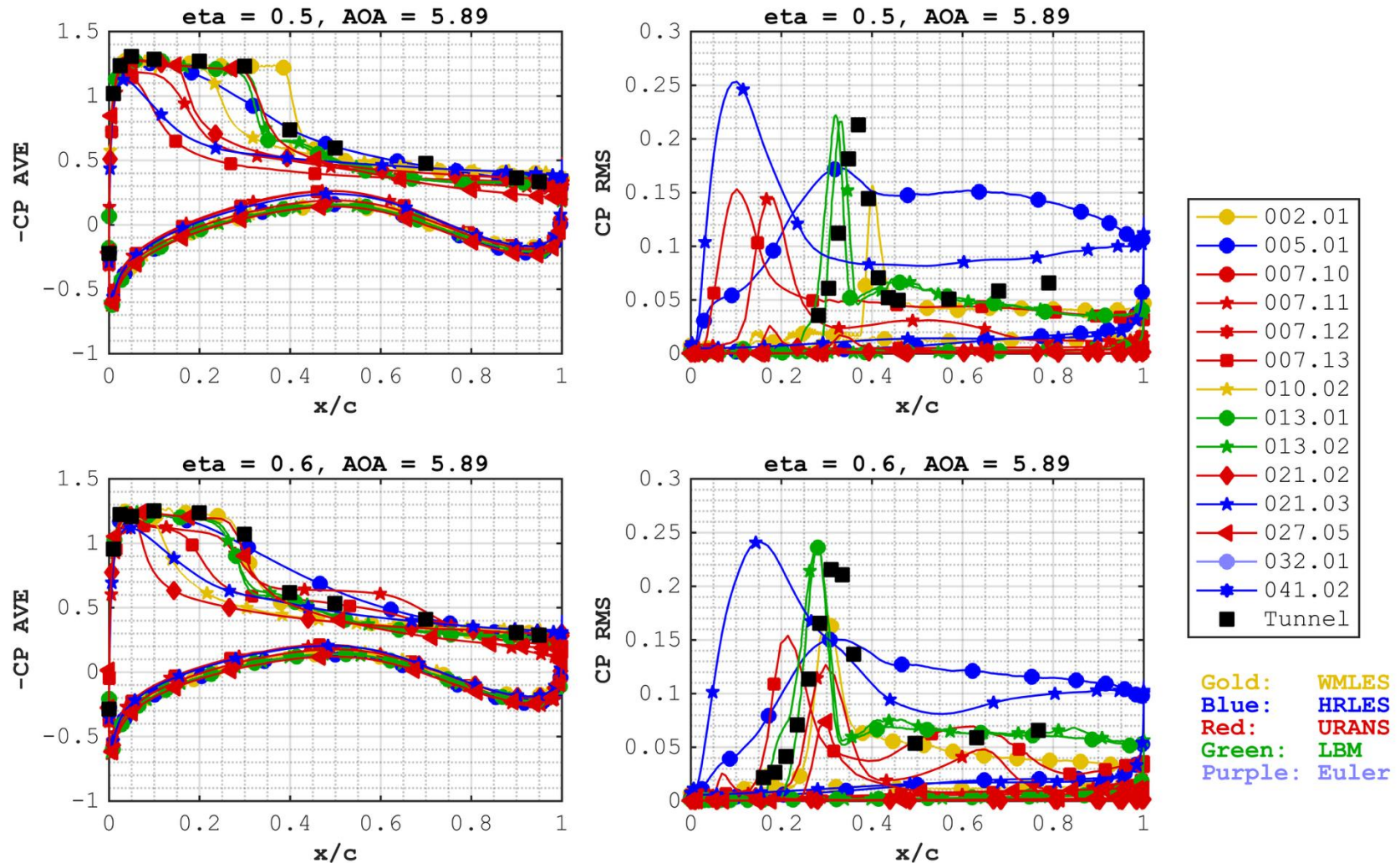


Temporal Formulation Effect – 4.84 deg

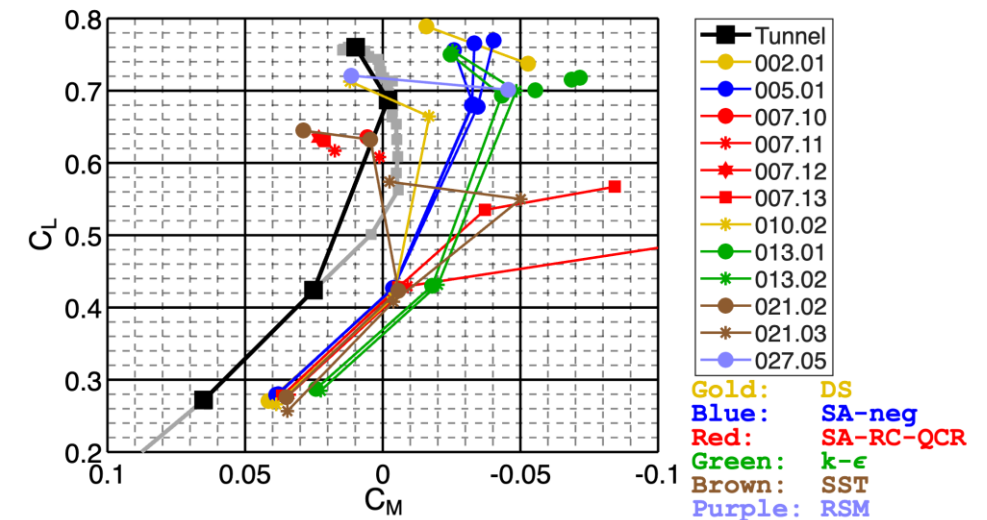
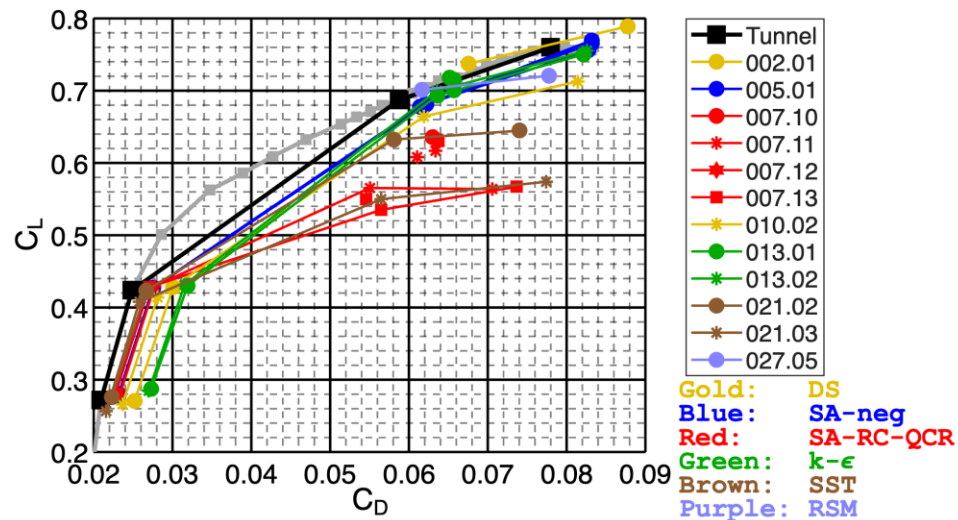
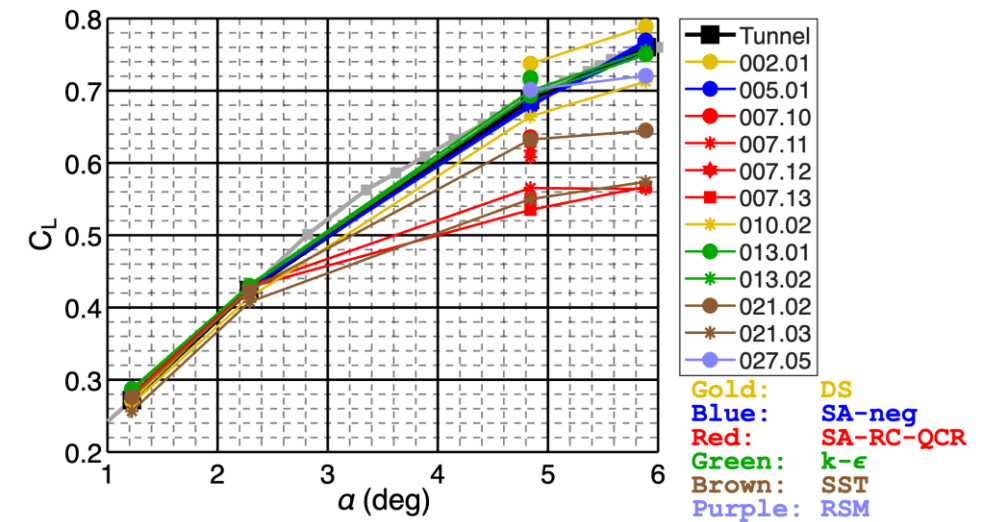


Temporal Formulation Effect – 5.89 deg

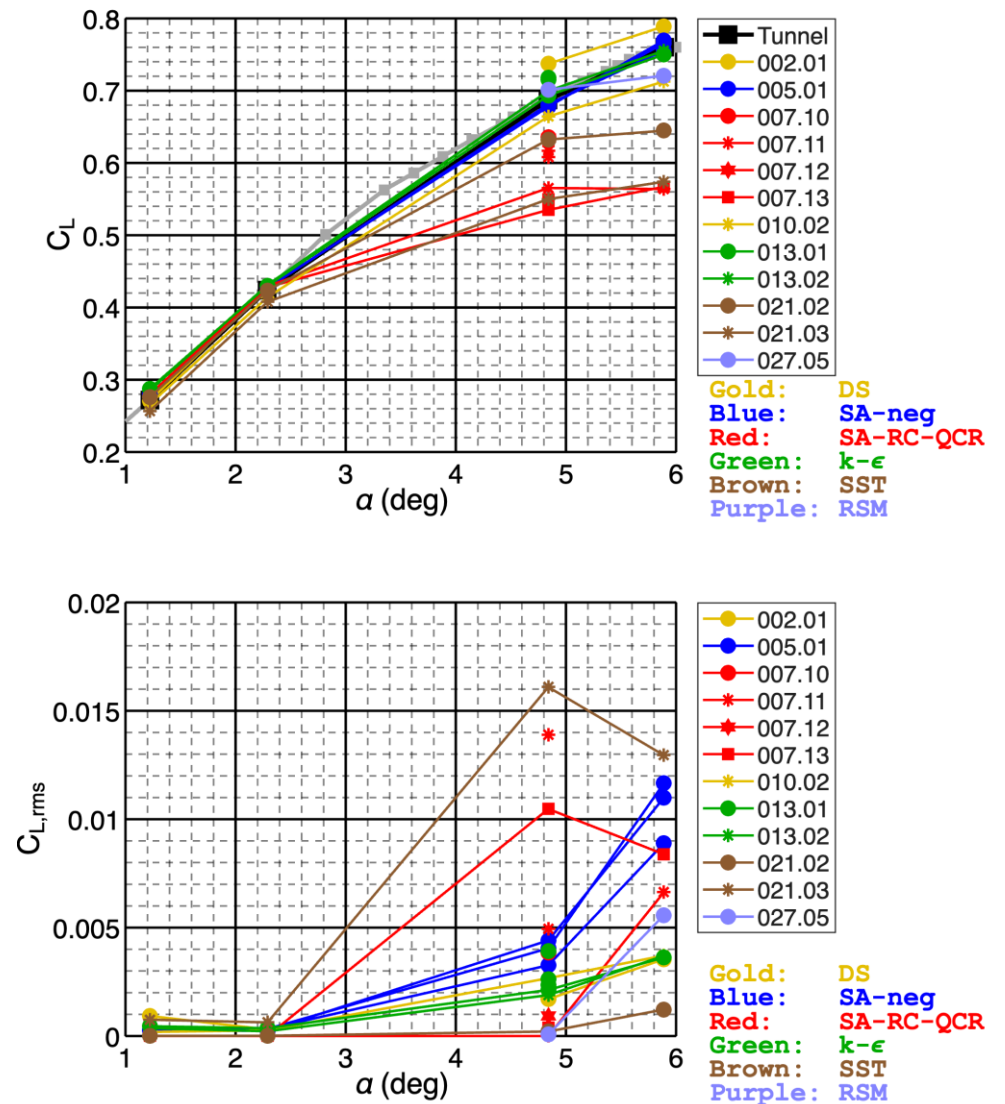
- Large code-to-code variations in the average shock locations and RMS spread, at 4.84 and 5.89 deg
- The sole LBM results are closest to the experimental data, particularly at 5.89 deg



- Submissions contain few scheme/turbulence
 - Hinders determining source of spread (is it sch
 - More overlap is desired
- The new SA-neg (shear-corrected) shows go

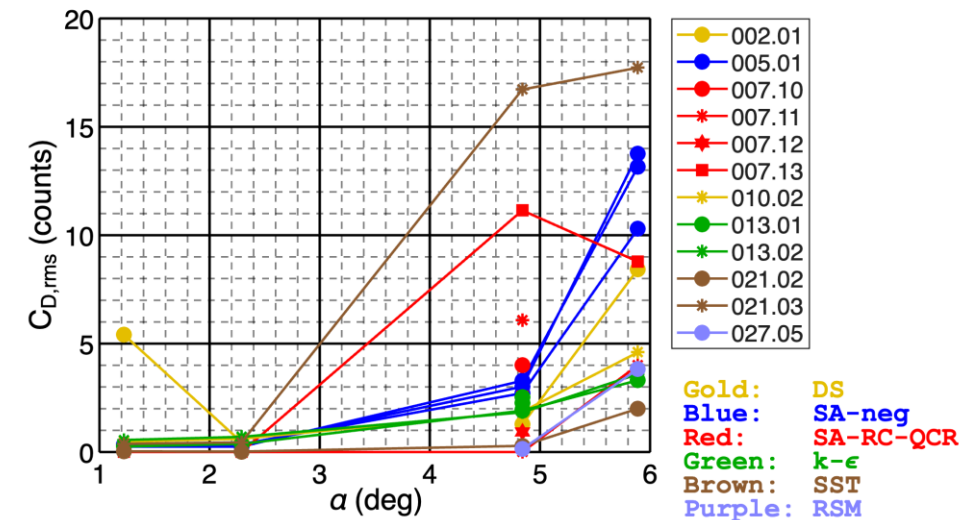
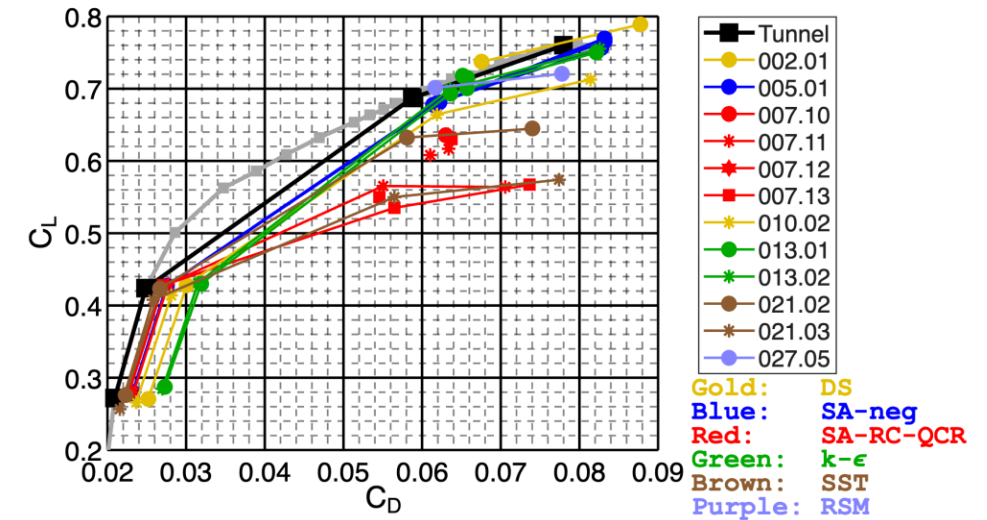


- **Low-alpha**
 - Minimal conclusive trends
- **High-alpha**
 - Drop in lift seen with URANS also observed with SA-RC-QCR (which of these is the cause?)
 - SST, URANS and HRLES, also shows reduced lift
 - SA-neg with Shear-Constrained LES yields good results



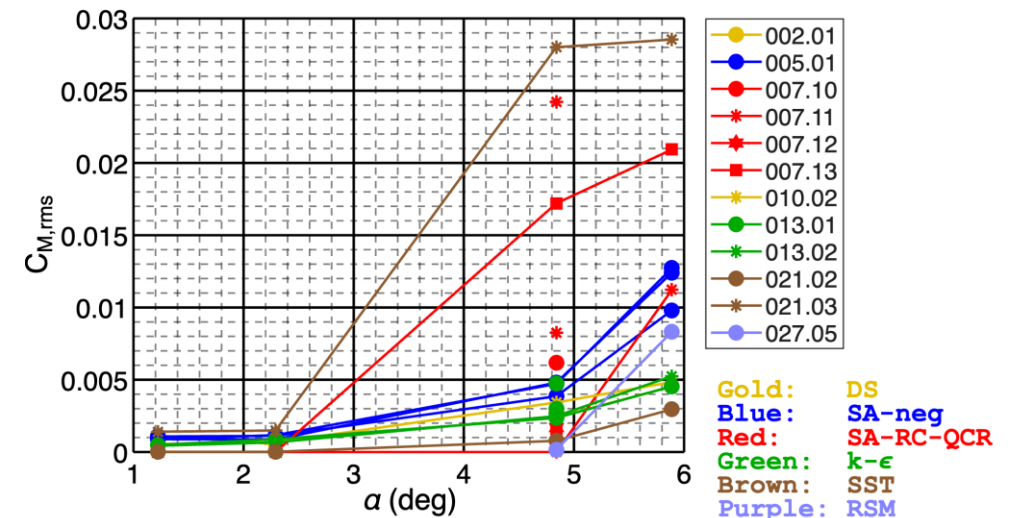
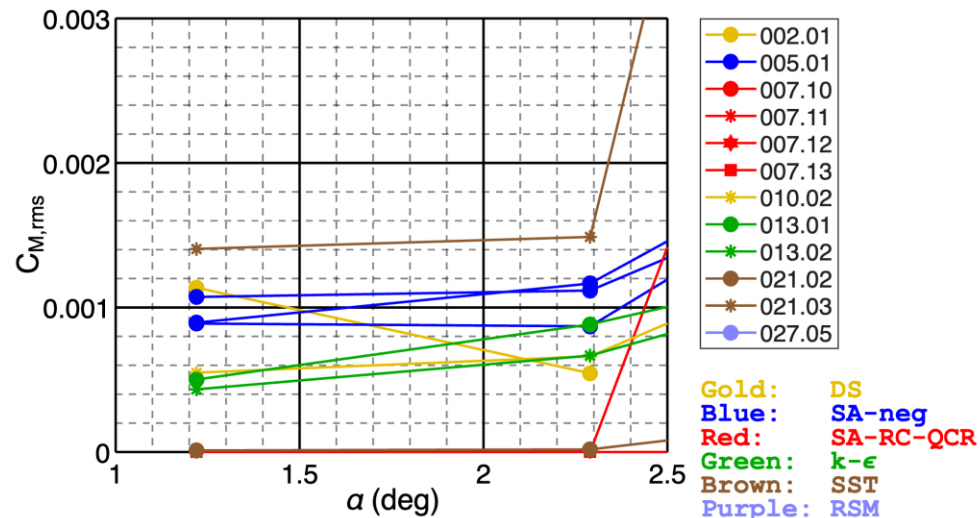
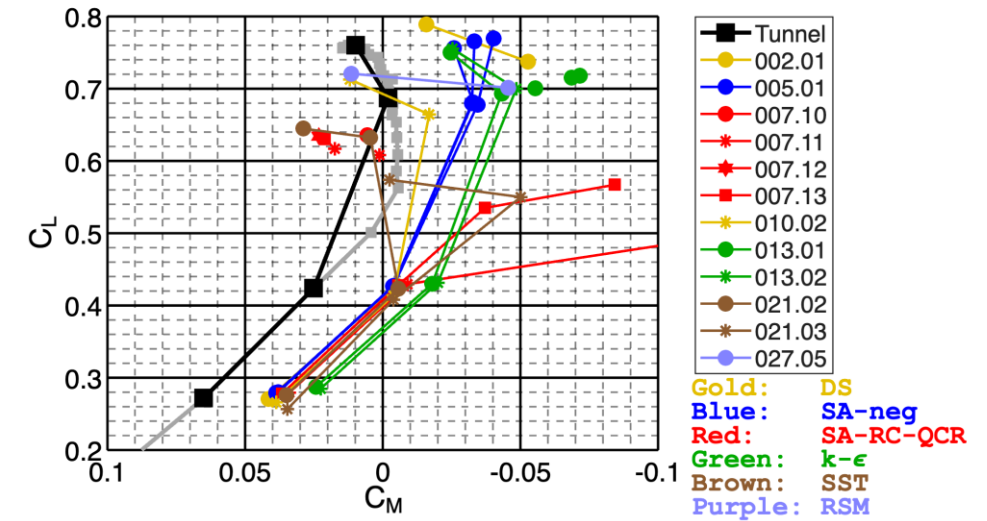
Turbulence Model Effect – Drag

- **Low-alpha**
 - WMLES/DS and LBM/k- ϵ are elevated;
is it scheme or turbulence model?
- **High-alpha**
 - SA-neg shows minimal spread with
different grid levels
 - SST indicates reduced lift and
increased $C_{D,rms}$

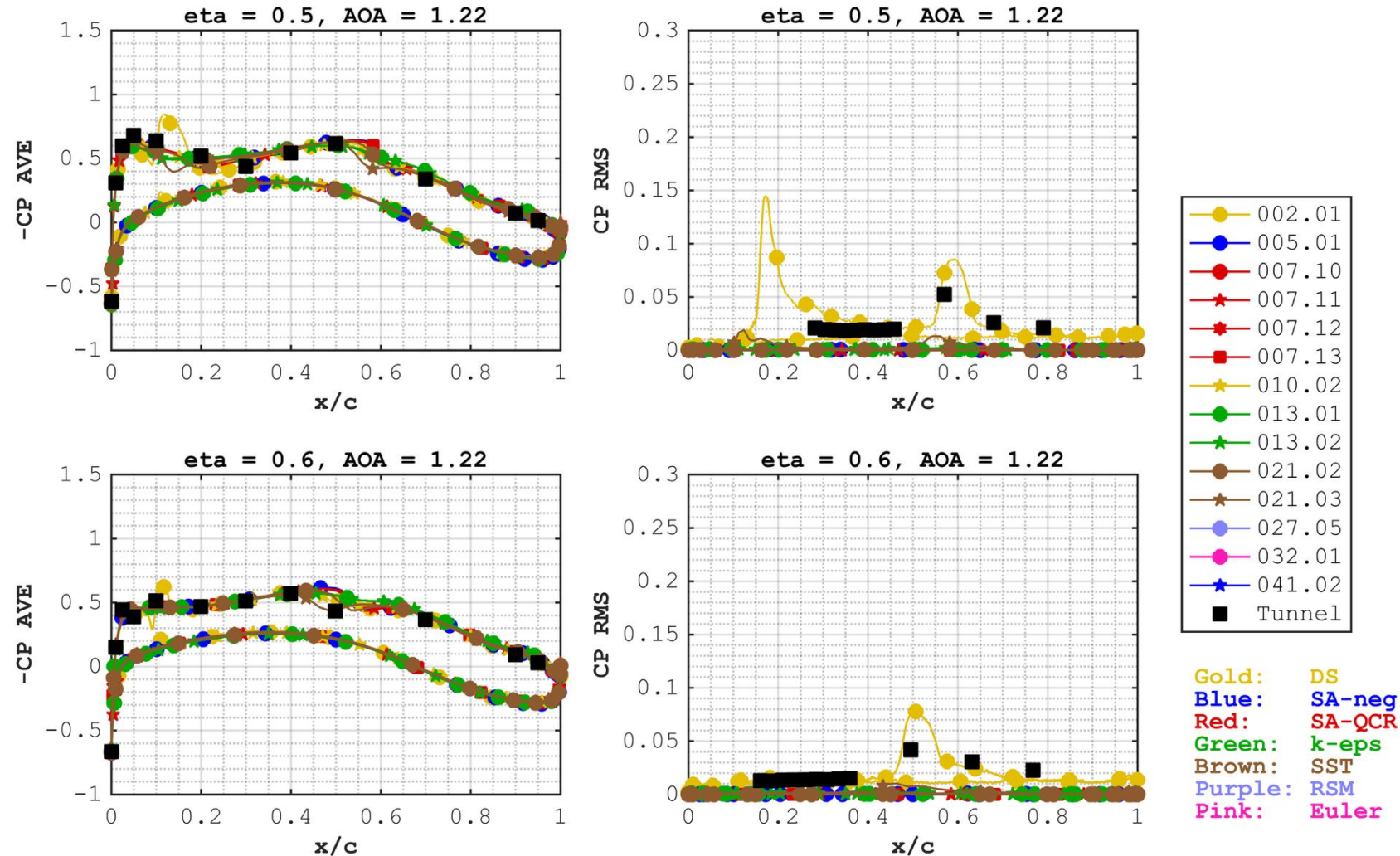


Turbulence Model Effect – Pitching Moment

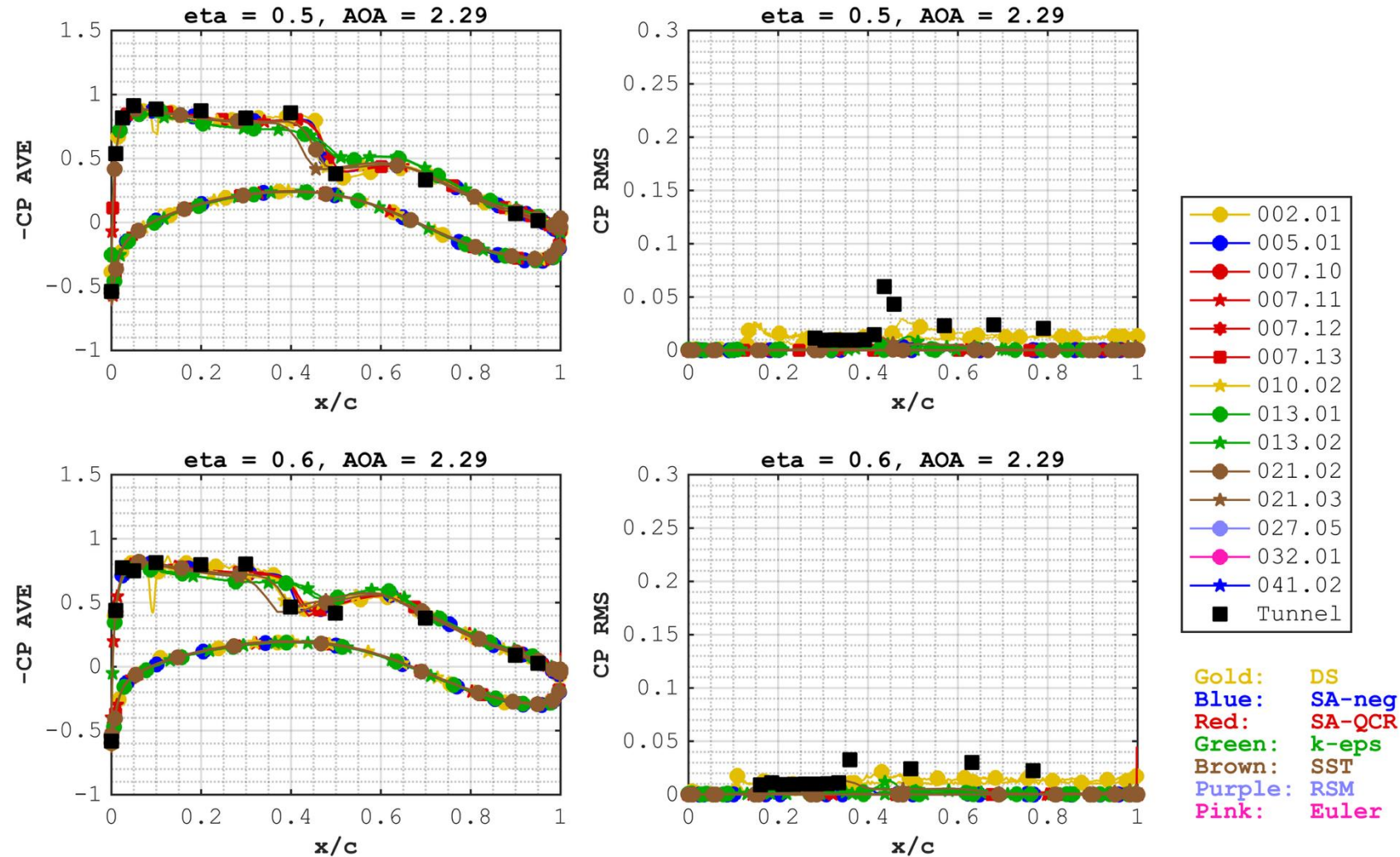
- **Low-alpha**
 - SST may yield excess $C_{M,rms}$ (truth unknown)
- **High-alpha**
 - SA-RC-QCR is sensitive to grid/time step
 - SST is sensitive to scheme
 - Questions remain regarding URANS and grid sensitivity



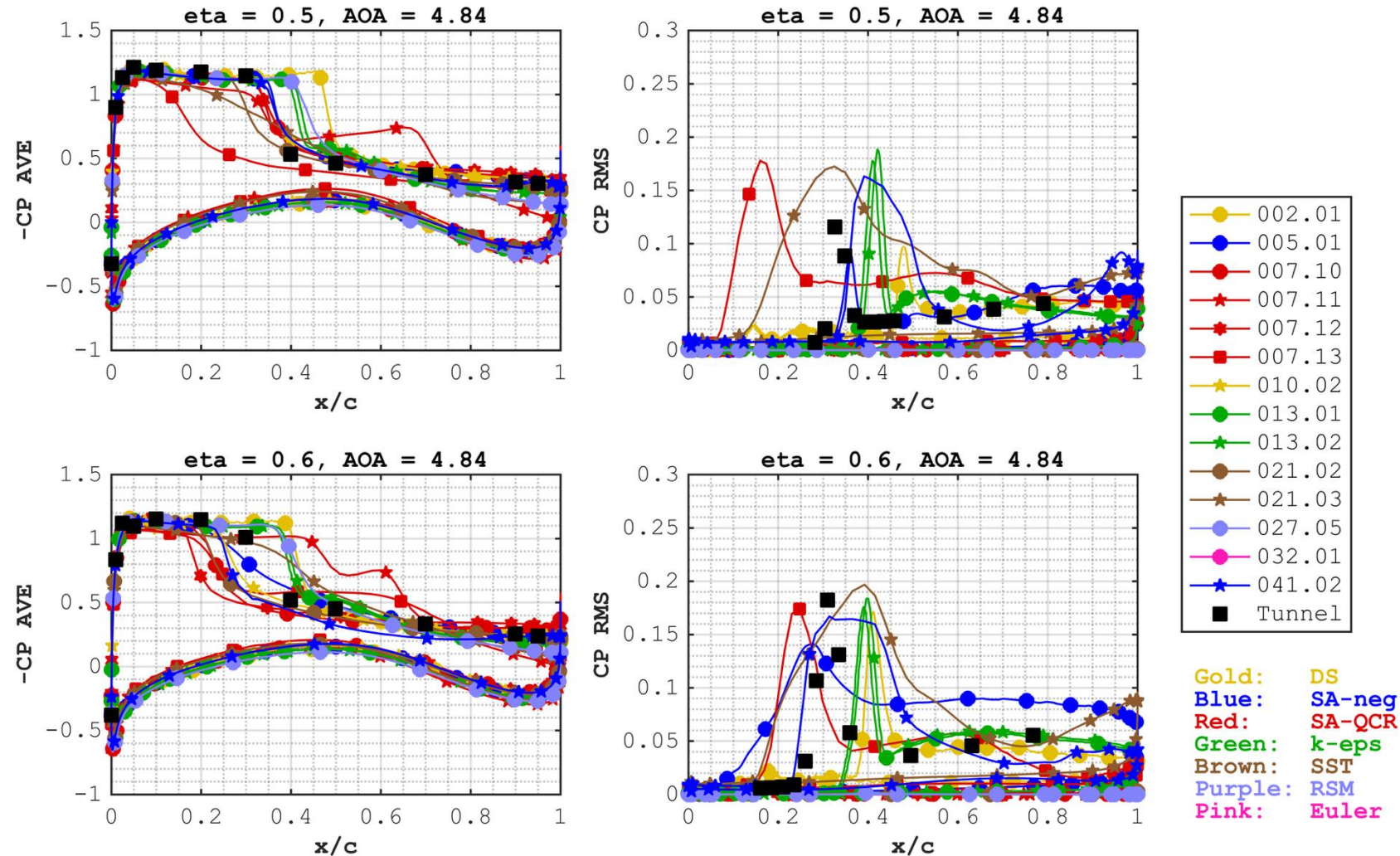
Turbulence Model Effect – 1.22 deg



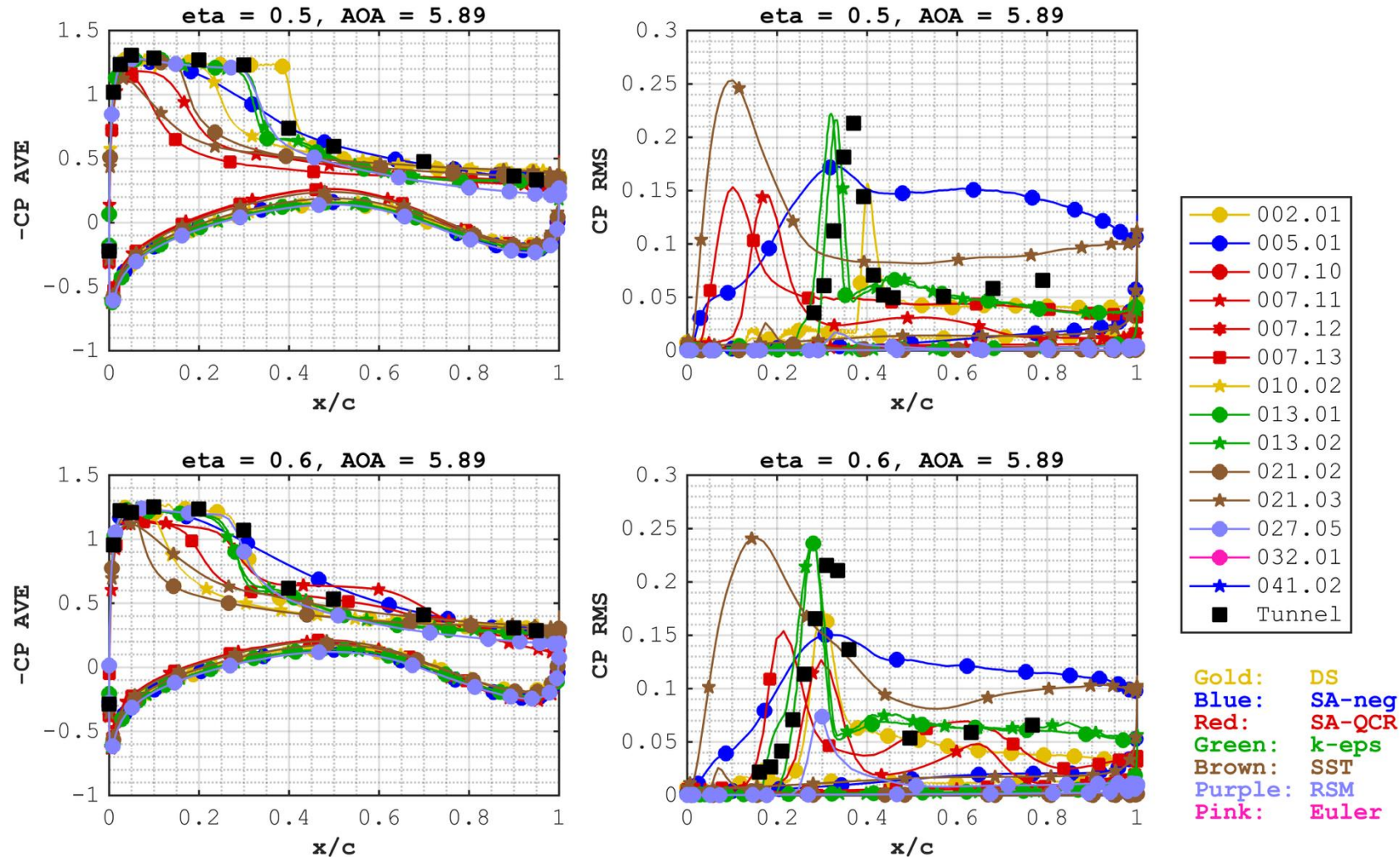
Turbulence Model Effect – 2.29 deg



Turbulence Model Effect – 4.84 deg

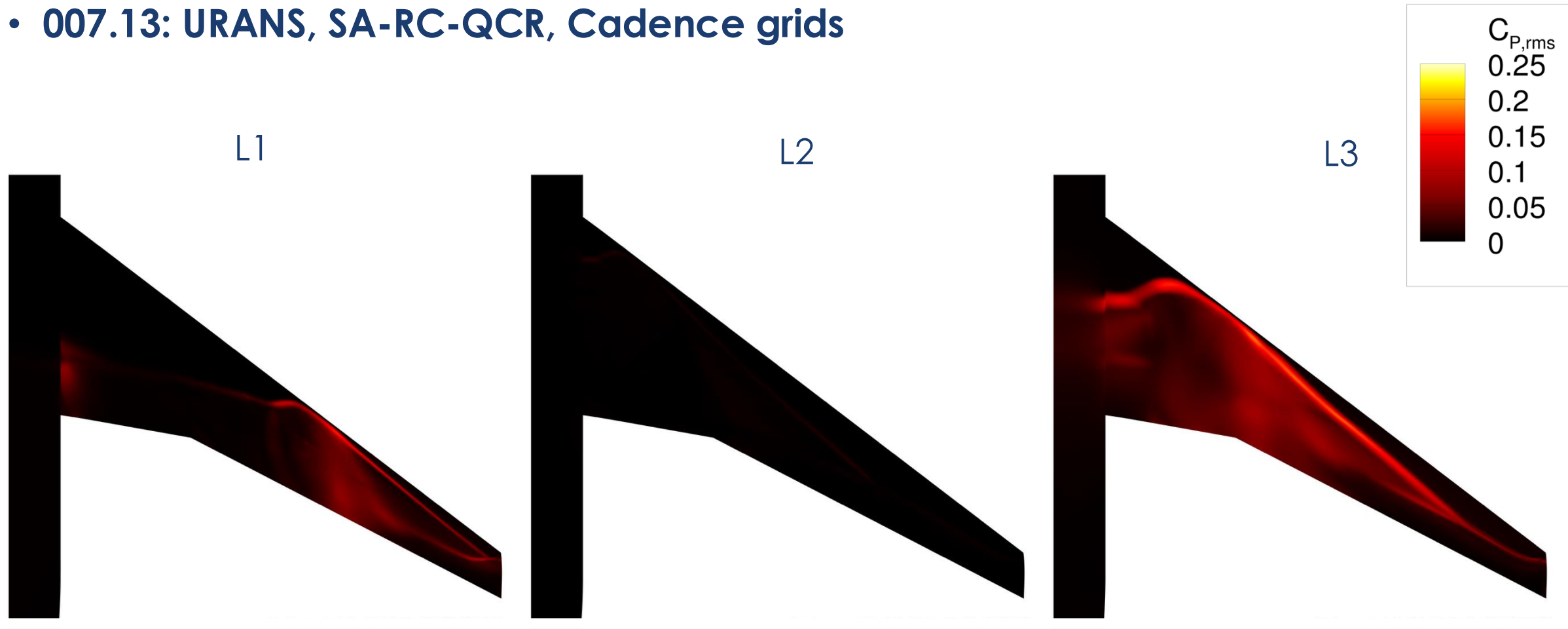


Turbulence Model Effect – 5.89 deg



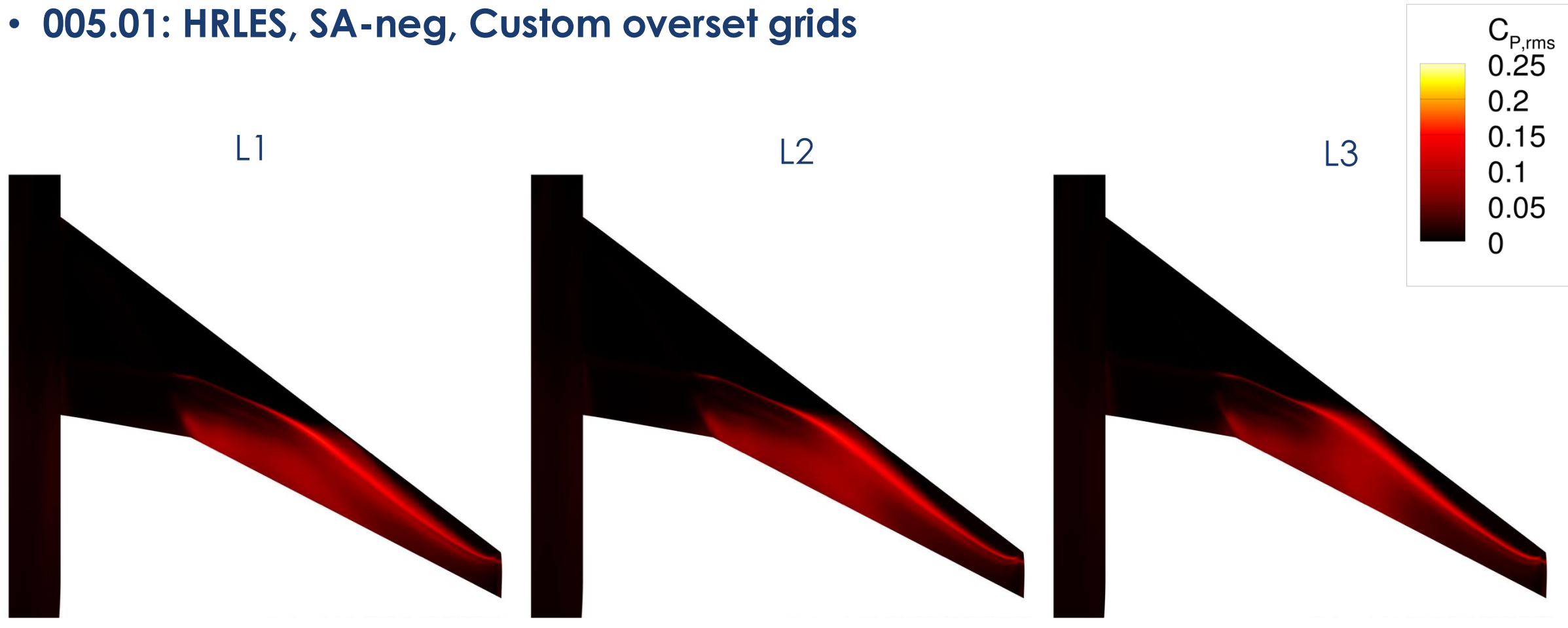
$C_{P,rms}$ Effect of Grid Density at 4.84 Deg (URANS)

- 007.13: URANS, SA-RC-QCR, Cadence grids



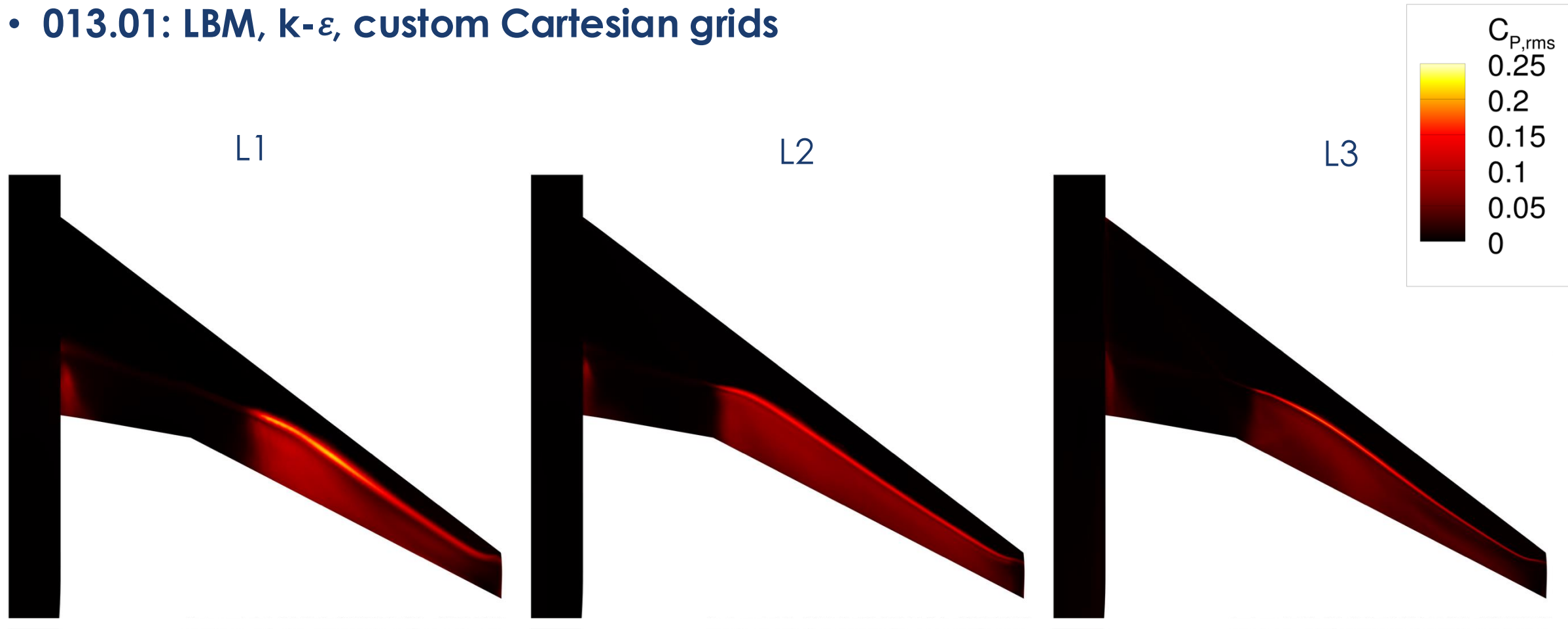
$C_{P,rms}$ Effect of Grid Density at 4.84 Deg (HRLES)

- 005.01: HRLES, SA-neg, Custom overset grids

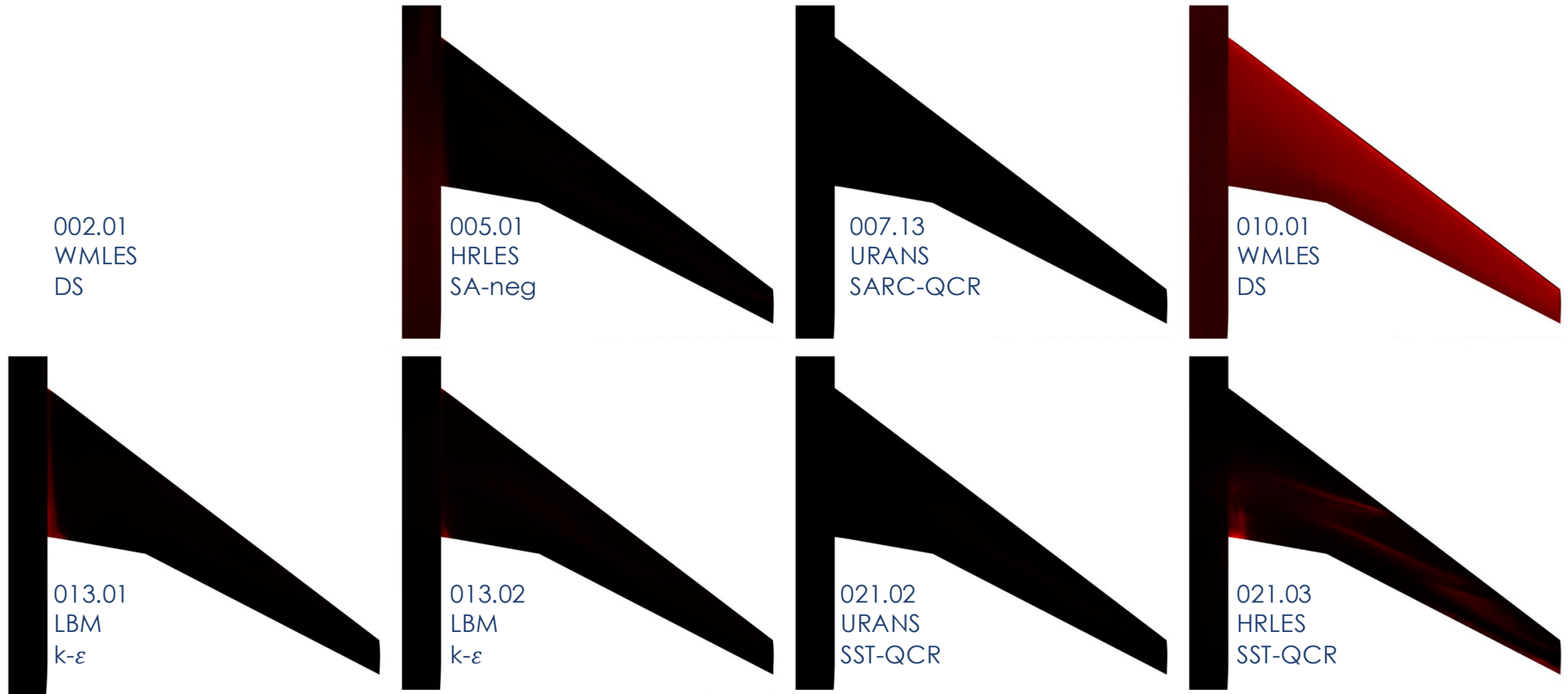


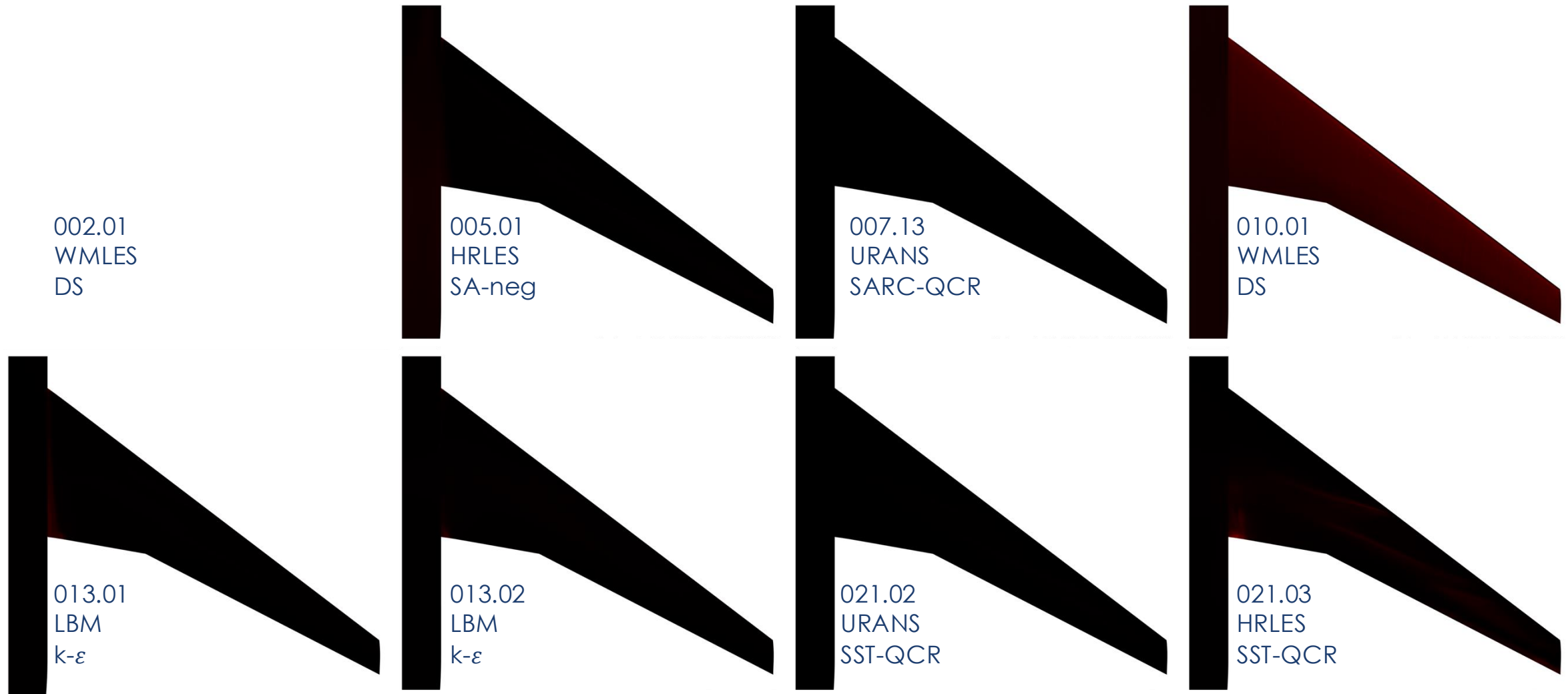
$C_{P,rms}$ Effect of Grid Density at 4.84 Deg (LBM)

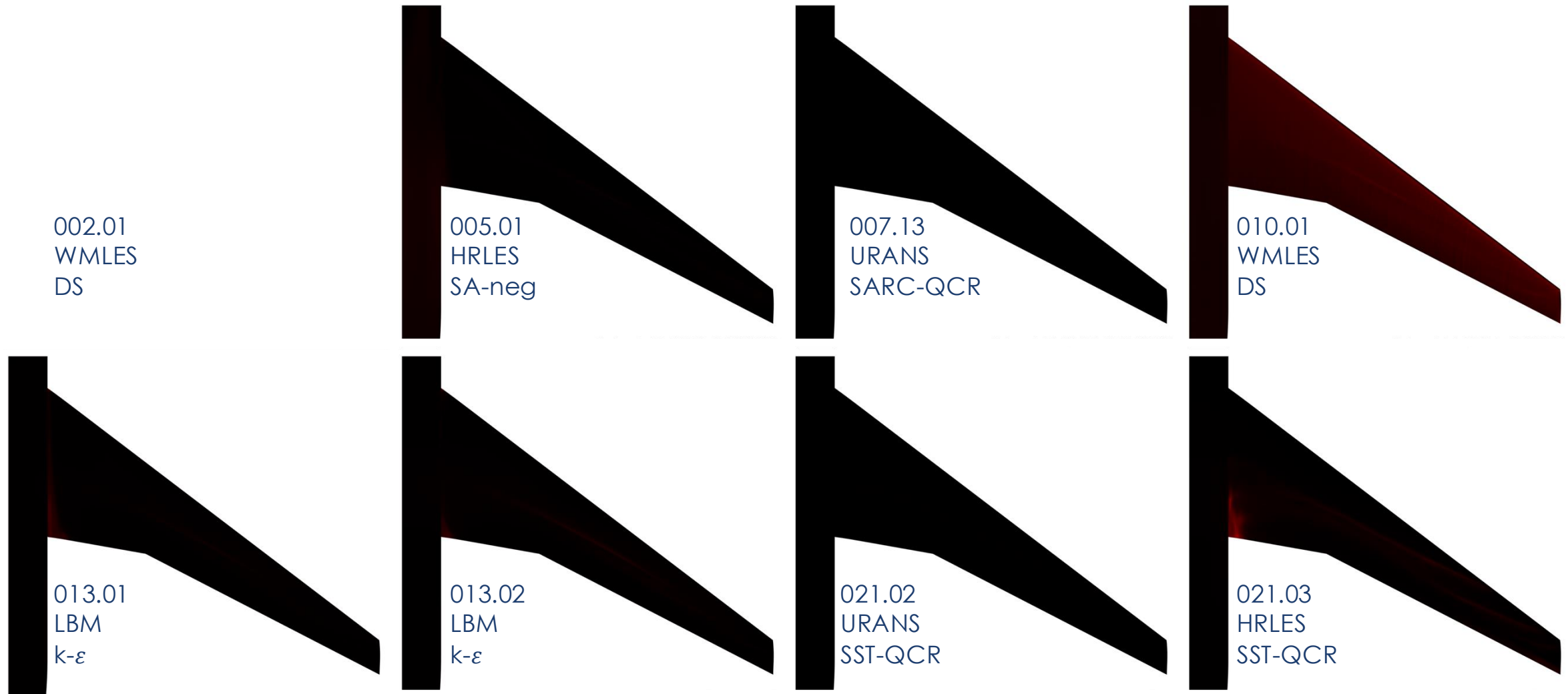
- 013.01: LBM, k- ϵ , custom Cartesian grids



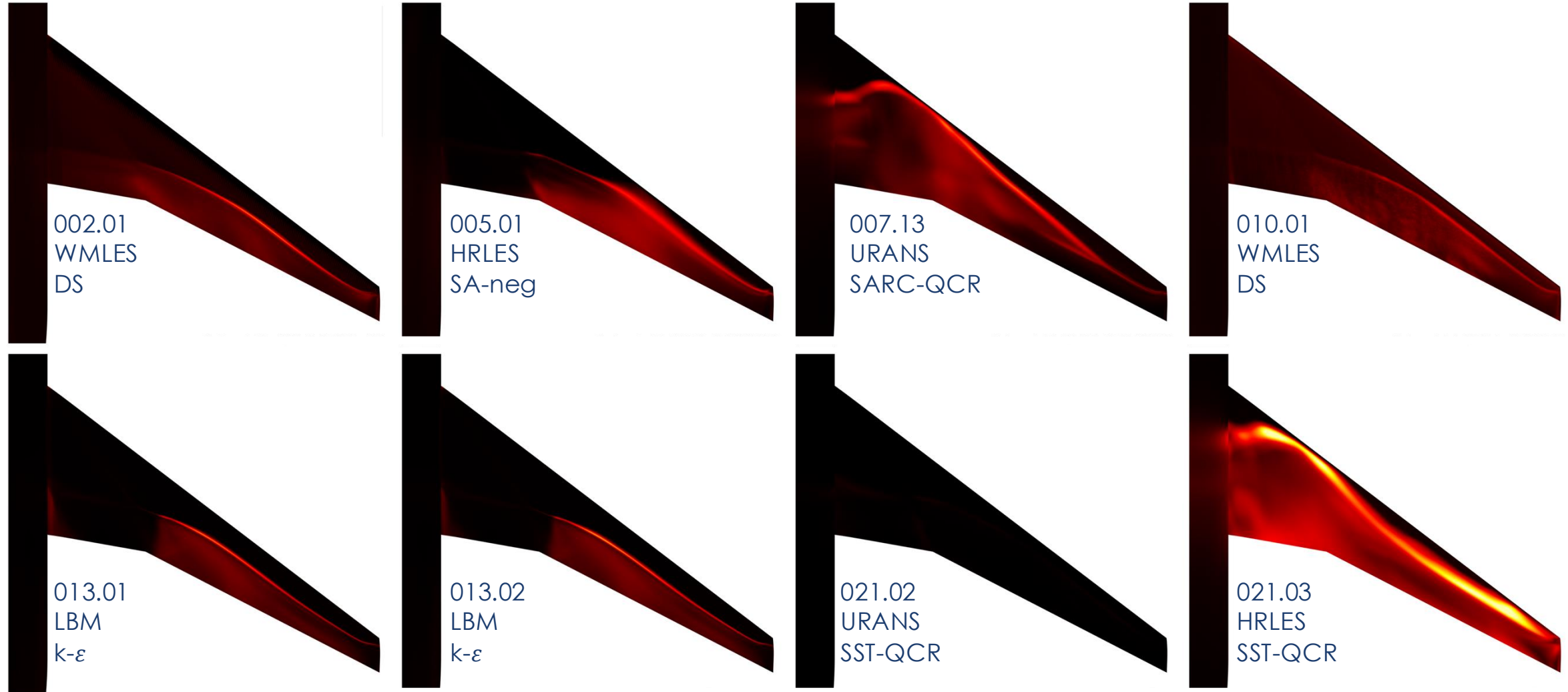
$C_{P,rms}$ (1.22 deg; Grid L3) – Expanded Contours



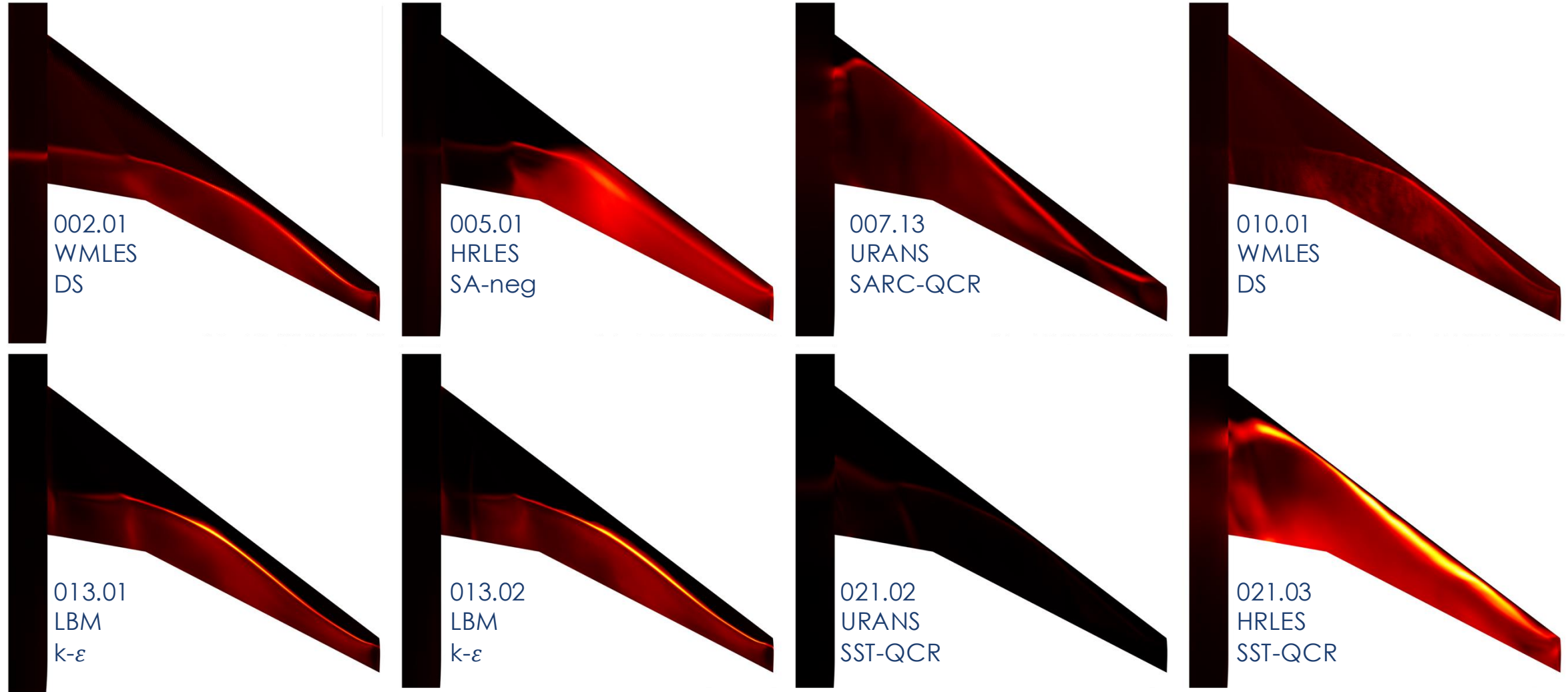


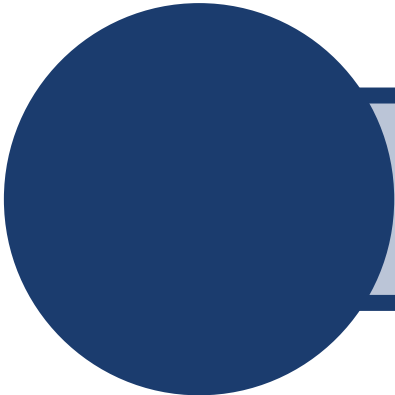


$C_{P,rms}$ (4.84 deg; Grid L3)



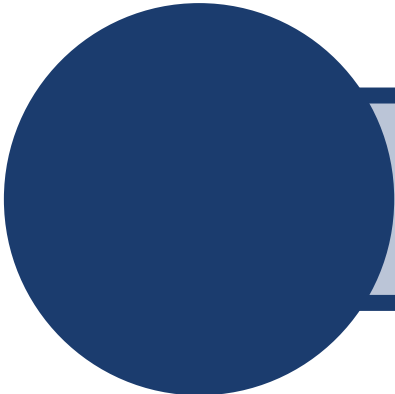
$C_{P,rms}$ (5.89 deg; Grid L3)



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Test Case 3

- **Difficult and expensive computational problem**
- **Some progress from four participants**
- **More to be discussed at SciTech**

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Open Questions and Desired Future Work

- Community is quite close to resolving some key questions
- Opportunities exist to round out the data set for SciTech and final paper
- You are encouraged to execute additional analysis as outlined
- Final data are due November 15, 2026 (hard stop)

- Ensemble results are understood quite well
- Surprising amount of scatter for an airfoil
- Grid convergence is difficult to achieve for unsteady schemes, even with attached flow
- Diminishing (or diminished) returns for additional simulations; focus upon Test Cases 2 and 3 would be more beneficial

1.) Is WMLES necessary? What can be solved with URANS and HRLES?

- *Execute additional HRLES simulations*
- *Constant turbulence model and scheme-appropriate grid, different schemes*

2.) Do any non-SA turbulence models predict the flowfield with sufficient accuracy?

- *Perform turbulence model sensitivities for a given scheme/approach at L3+*

3.) Is URANS a dead end? Can HRLES be sufficient?

- *Execute URANS and HRLES with finer grids (L4+) for Cadence and Helden grids*
- *Evaluate and increase subiterative convergence*

4.) What grid convergence metrics can we discern?

- *Execute simulations on L2 and L4*

5.) Can point-matched data shed light on the high-alpha conditions?

- *Execute simulations at $\alpha = 2.29$ deg*

1.) Do we have any quantitative data of reasonable confidence?

- *Work-in-progress data are encouraged*
- *For this test case only, data can be published without your Participant ID if desired to encourage work-in-progress submissions*

2.) What lessons have been learned so far?

- *Significant information can be captured without quantitative results*
- *Have code or method struggles been identified?*
- *What improvements must be made to grid-movement schemes? Can simulations start from a jig shape?*

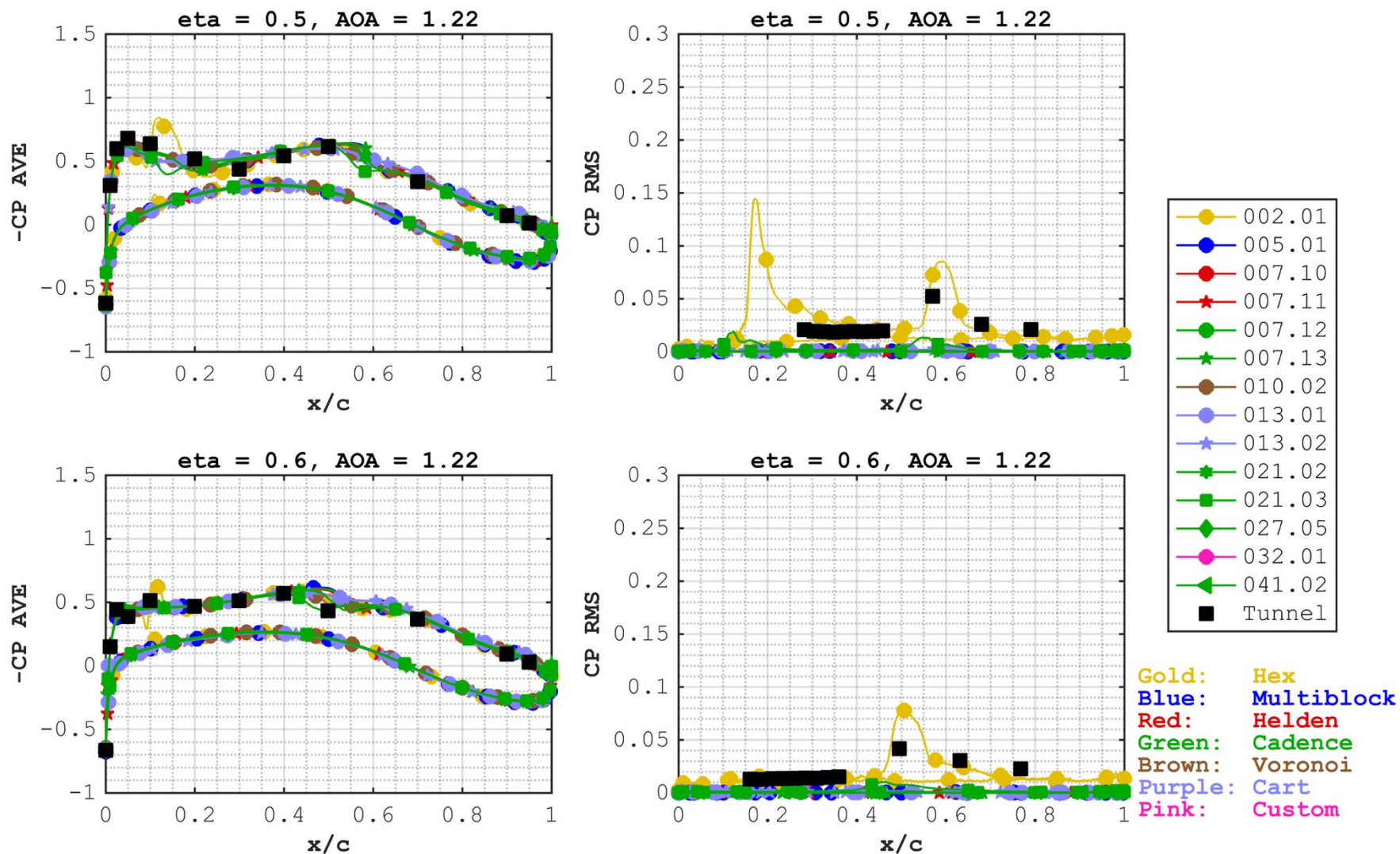
3.) Can you provide any key images depicting your results, grid, troubleshooting process, etc.?

- **Desired high Reynolds number experimental data set**
 - Physics at 20M or 30M are very different than 1.5M or 2M
 - uPSP, F&M time histories, and dynamic deformation metrics
- **Robust grid sensitivity studies**
 - Grid convergence for scale-resolving techniques for very fine grids
 - Grid movement improvements, especially adaptation
- **Effect of nacelle/pylon (and strake?)**
- **Dynamic vehicle considerations**
- **Side-of-body physics**

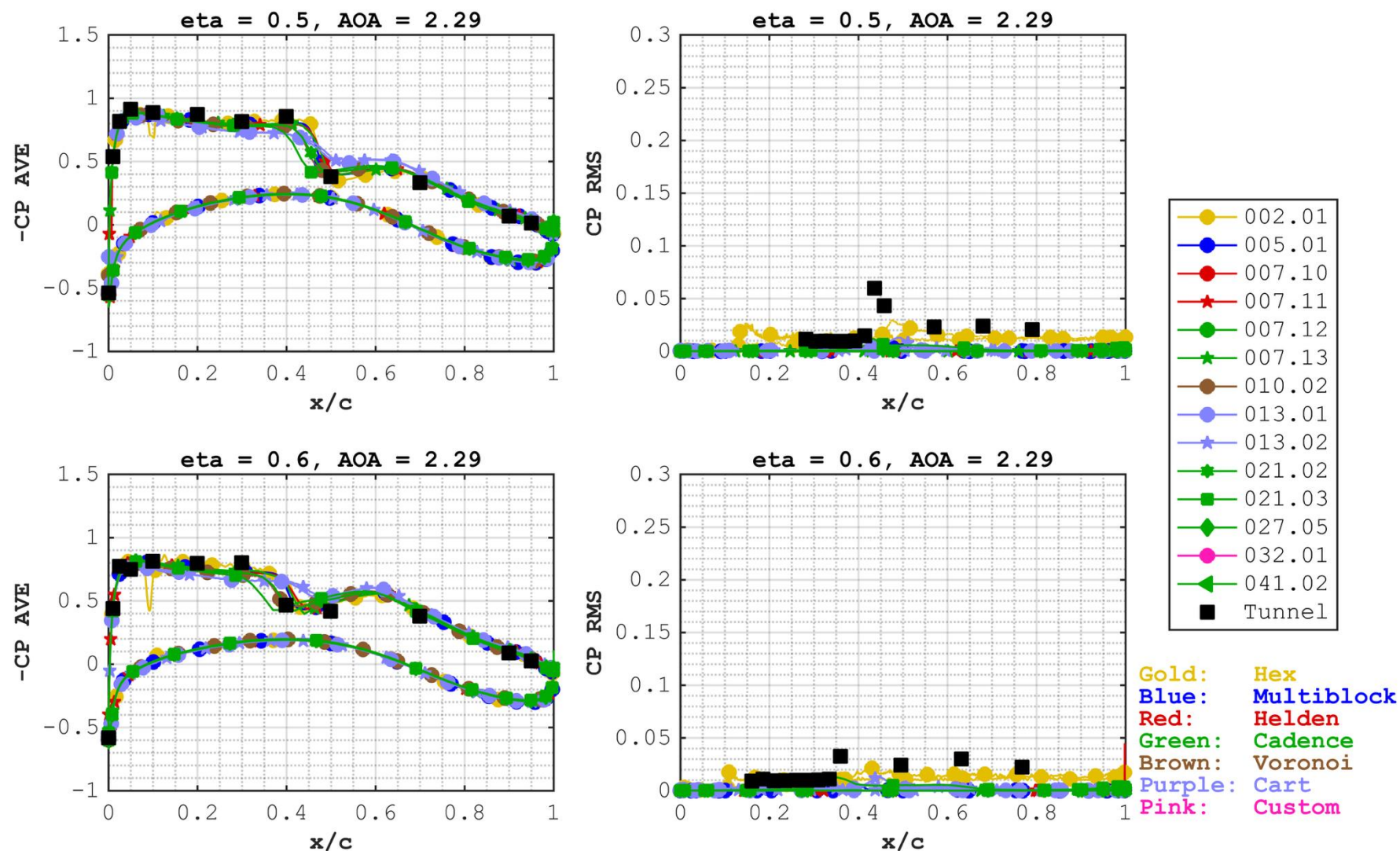
- **Appreciation is extended to**
 - Craig Hunter, NASA Langley Research Center (Emeritus)
 - Michael Lee, NASA Langley Research Center
 - Chris Rumsey, NASA Langley Research Center (Emeritus)
 - Ethan Vogel, NASA Langley Research Center



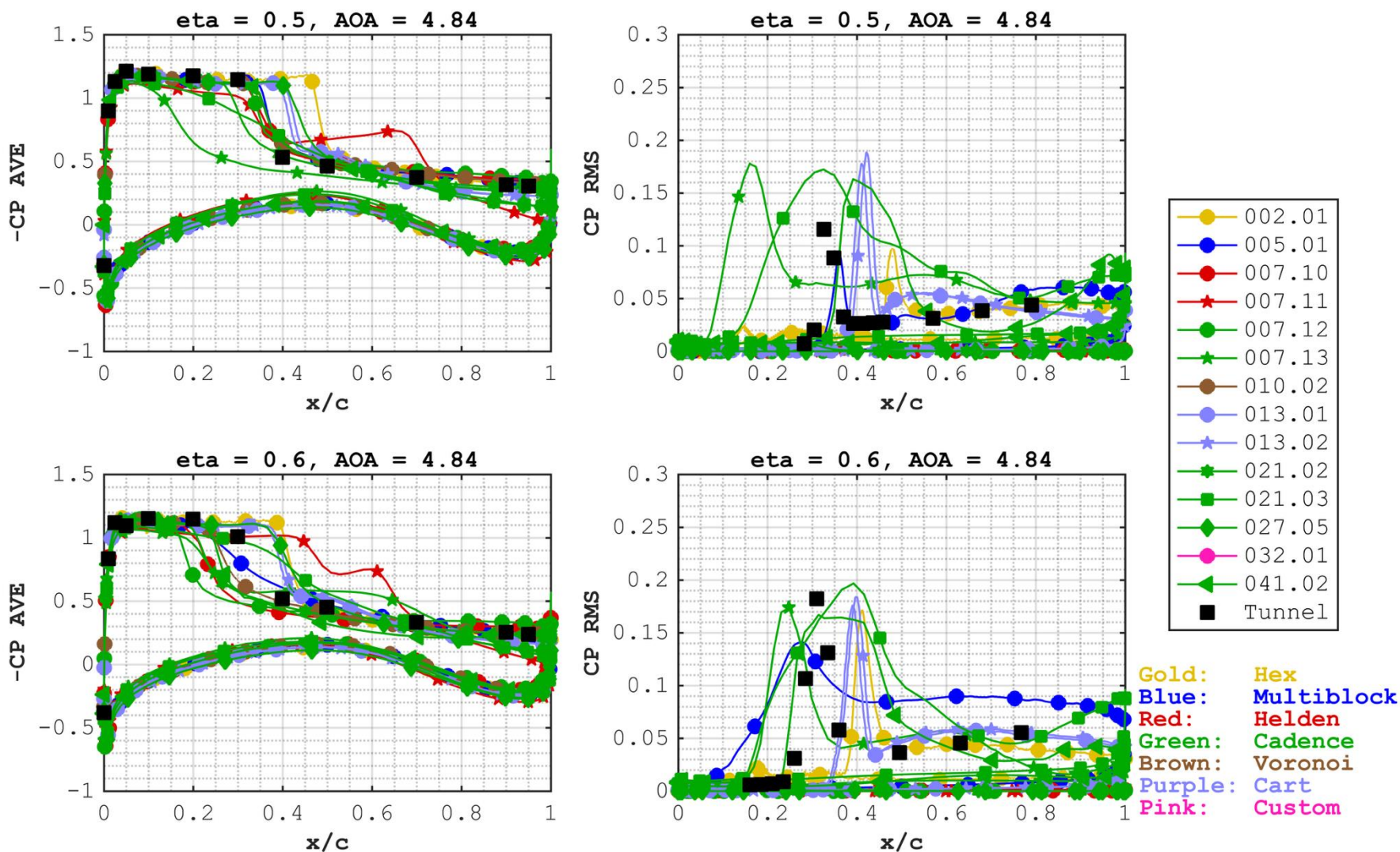
Grid Type Effect – Unsteady Cuts, AoA = 1.22 deg.



Grid Type Effect – Unsteady Cuts, AoA = 2.29 deg.



Grid Type Effect – Unsteady Cuts, AoA = 4.84 deg.



Grid Type Effect – Unsteady Cuts, AoA = 5.89 deg.

